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Input sensing slidable structure of flexible display module and electronic device including the same

Abstract

An electronic device includes a display module including a first region and a second region which extends from the first region and tactically senses an input via a conductive layer, a first structure facing the display module, a second structure slidably connected to the first structure, and slidable together with the display module along the first structure, and a back cover forming a rear surface of the electronic device and including a window region, and the conductive layer which is on the window region. The second structure which slides towards and away the first structure respectively closes and opens the electronic device, the electronic device which is closed defines the front surface including the first region, together with the second region tactically exposed at the window region, and the electronic device which is open defines the front surface including the first region and a portion of the second region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Bypass Continuation Application of International Application No. PCT/KR2021/007489 designating the United States, filed on Jun. 15, 2021 in the Korean Intellectual Property Receiving Office, and claiming priority to Korean Patent Application No. 10-2020-0073042, filed on Jun. 16, 2020, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
Field
(1) Various embodiments of the disclosure described herein relate to an electronic device including a flexible display module.
Description of the Related Art
(2) An electronic device may include a flexible display module. The electronic device may expand a display region visually exposed on an outer surface of the electronic device. For example, the flexible display module may be disposed in a curved, foldable, or rollable form in the electronic device.

DISCLOSURE

Technical Problem

(3) Recently, various forms of electronic devices have been developed to secure an expanded display region without affecting portability. For example, the electronic devices may include a slide type electronic device in which a first structure (e.g., a first housing) and a second structure (e.g., a second housing) slide relative to each other, or a foldable electronic device in which a first housing and a second housing are disposed to be folded or unfolded.

(4) In the slide type electronic device among the various forms of electronic devices, a display region exposed in a front direction of the electronic device may be expanded as a flexible display is withdrawn by sliding of the first structure (e.g., the first housing) and the second structure (e.g., the second housing). In this case, at least a portion of the flexible display may be disposed to face a rear direction of the electronic device as the flexible display is rolled in a closed state in which the first structure and the second structure overlap each other.

(5) Various embodiments of the disclosure provide an electronic device capable of visually exposing at least a portion of a flexible display in a rear direction of the electronic device in a closed state of the electronic device. In addition, various embodiments of the disclosure provide an electronic device capable of high-sensitivity touch input on a rear surface of an electronic device in a closed state of the electronic device.

SUMMARY

(6) An electronic device according to an embodiment of the disclosure includes a first structure, a second structure that is connected to the first structure so as to slide relative to the first structure and that surrounds at least a portion of the first structure, a display module that is disposed on the second structure and that is movable together with the second structure relative to the first structure, and a back cover that forms at least a portion of an outer surface of the electronic device and that is disposed to face the first structure. The display module includes a first region and a second region that extends from the first region, and the first region or the second region is formed to be at least partially flexible. The back cover includes a window region having at least a partial region formed of a transparent or translucent material and a conductive layer disposed on at least a portion of the window region to face toward the first structure. The electronic device includes a first state in which the first region forms a front surface of the electronic device and at least a portion of the second region is disposed between the first structure and the back cover and visually exposed on a rear surface of the electronic device through the window region and a second state in which at least a portion of the second region forms the front surface of the electronic device together with the first region. In the first state, at least a portion of the second region is configured to receive a touch input from the outside through the window region and the conductive layer.

(7) An electronic device according to an embodiment of the disclosure includes a case including a support member, a first side member, and a second side member, the first side member and the second side member being disposed on opposite end portions of the support member, a first structure having at least a portion surrounded by the case, a second structure connected with the first structure so as to slide relative to the first structure, a display module disposed on the second structure so as to be movable together with the second structure relative to the first structure and formed to be at least partially flexible, the display module including a first region and a second region that extends from the first region, and a back cover that forms at least a portion of a rear surface of the electronic device and that is disposed to surround the support member, the back cover including a window region formed to be transparent and a conductive layer disposed on at least a portion of the window region. The electronic device is changed between a first state and a second state as the second structure slides. In the first state, the first region forms a front surface of the electronic device, and at least a portion of the second region is visually exposed on the rear surface of the electronic device through the window region. In the second state, at least a portion of the second region forms the front surface of the electronic device together with the first region. In

the first state, at least a portion of the second region is located to face the window region and the conductive layer and configured to receive a touch input from the outside through the window region and the conductive layer.
Advantageous Effects
(8) The electronic device according to the various embodiments of the disclosure may have an exposed region in the back cover such that at least a portion of the flexible display module is visually exposed through the back cover. Accordingly, the front surface and the rear surface of the electronic device may be used as a display region.
(9) Furthermore, the electronic device according to the various embodiments of the disclosure may have a conductive material disposed or formed on at least a portion of the back cover, thereby improving the sensitivity of a touch input through the back cover.
(10) In addition, the disclosure may provide various effects that are directly or indirectly recognized.

Description

DESCRIPTION OF DRAWINGS
(1) The above and other advantages and features of this disclosure will become more apparent by describing in further detail embodiments thereof with reference to the accompanying drawings, in which
(2) FIGS. **1**A and **1**B are views illustrating a first state of an electronic device according to an embodiment.
(3) FIGS. **2**A and **2**B are views illustrating a second state of the electronic device according to an embodiment.
(4) FIG. **3** is an exploded perspective view of the electronic device according to an embodiment.
(5) FIGS. **4**A and **4**B are cross-sectional views of the electronic device according to an embodiment.
(6) FIG. **5** is a sectional view illustrating a coupling relationship between a second structure, a first roller member, a second roller member, and a belt member of the electronic device according to an embodiment.
(7) FIGS. **6**A and **6**B illustrate an operation in which a touch is input through a back cover in the first state of the electronic device according to an embodiment.
(8) FIGS. **7**A to **7**C illustrate the back cover of the electronic device according to an embodiment.
(9) FIGS. **8**A to **8**C illustrate the back cover of the electronic device according to an embodiment.
(10) FIGS. **9**A to **9**G illustrate an electrode pattern of a conductive layer of the electronic device according to an embodiment.
(11) FIGS. **10**A to **10**E illustrate a display module of the electronic device according to an embodiment.
(12) FIG. **11** is a block diagram illustrating an electronic device in a network environment according to an embodiment.
(13) FIG. **12** is a block diagram illustrating the display device according to an embodiment.
(14) With regard to description of the drawings, identical or similar reference numerals may be used to refer to identical or similar components.
DETAILED DESCRIPTION
(15) Hereinafter, various embodiments of the disclosure may be described with reference to accompanying drawings. Accordingly, those of ordinary skill in the art will recognize that modification, equivalent, and/or alternative on the various embodiments described herein can be variously made without departing from the scope and spirit of the disclosure.
(16) FIGS. **1**A and **1**B (otherwise referred to as FIG. **1**) are views illustrating a first state of an

electronic device according to an embodiment. FIGS. **2**A and **2**B (otherwise referred to as FIG. **2**) are views illustrating a second state of the electronic device according to an embodiment.

(17) FIGS. **1**A and **1**B are views respectively illustrating a front surface and a rear surface of the electronic device when the electronic device is in the first state. FIGS. **2**A and **2**B are views respectively illustrating the front surface and the rear surface of the electronic device when the electronic device is in the second state.

(18) Referring to FIGS. **1** and **2**, the electronic device **100** according to an embodiment may include a case **110**, a second structure **140**, a display module **150**, and a back cover **160**. According to an embodiment, the electronic device **100** may include the first state (e.g., the state of FIG. **1**) and the second state (e.g., the state of FIG. **2**). For example, the first state and the second state of the electronic device **100** may be determined depending on the position of the second structure **140** relative to the case **110**, and the electronic device **100** may be configured to be changed (e.g. changeable) between the first state and the second state by a manual operation such as a user operation or an automatic or electrical operation such as a mechanical operation.

(19) In an embodiment of the disclosure, the first state of the electronic device **100** may refer to a closed state in which the electronic device **100** is closed (e.g., the electronic device **100** which is closed) such that at least a portion (e.g., a first peripheral portion **143**) of the second structure **140** is brought into contact with the case **110**. That is, the electronic device **100** which is closed disposes the first peripheral portion **143** in contact with the case **110**. The second state of the electronic device **100** may refer to an opened state in which the electronic device **100** is opened such that at least a portion (e.g., the first peripheral portion **143**) of the second structure **140** is spaced apart from the case **110**. That is the electronic device **100** which is open disposes the first peripheral portion **143** spaced apart from (e.g., non-contacting) the case **110**. As illustrated, a state (the state of FIG. **1**) in which a portion (e.g., a second peripheral portion **144** or a third peripheral portion **145**) of the second structure **140** is inserted into (e.g., inside) side members **110-1** and **110-2** of the case **110** may be defined as the first state, and a state (the state of FIG. **2**) in which a portion (e.g., the second peripheral portion **144** or the third peripheral portion **145**) of the second structure **140** is withdrawn from (e.g., extended outside) the side members **110-1** and **110-2** of the case **110** may be defined as the second state.

(20) In an embodiment of the disclosure, a surface facing a direction substantially the same as a direction that at least a portion (e.g., a first region **151**) of the display module **150** located outside the electronic device **100** faces may be defined as the front surface of the electronic device **100**, and a surface facing away from the front surface may be defined as the rear surface of the electronic device **100**. A surface surrounding a space between the front surface and the rear surface may be defined as a side surface of the electronic device **100**. For example, the front surface of the electronic device **100** may refer to surfaces that form the exterior of the electronic device **100** when a portion (e.g., the first region **151**) of the display module **150** located outside the electronic device **100** is viewed, and the rear surface of the electronic device **100** may refer to surfaces that form the exterior of the electronic device **100** when the back cover **160** is viewed. An outer surface of the electronic device **100** that substantially faces the +Z-axis direction may be construed as the front surface of the electronic device **100**, and an outer surface of the electronic device **100** that substantially faces the −Z-axis direction may be construed as the rear surface of the electronic device **100**.

(21) In an embodiment, the case **110** may form at least a portion of the side surface of the electronic device **100**, and the electronic device **100** may be changed between the first state and the second state as the second structure **140** and the display module **150** slide relative to the case **110** in opposite directions (e.g., a first direction D**1** and a second direction D**2**). The second structure **140** and the display module **150** may be slidable relative to the case **110**. A sliding direction of the second structure **140** and the display module **150** may be defined along the first direction D**1** and the second direction D**2**. The second structure **140** and the display module **150** may be slidable

together with each other, relative to the case **110**.

(22) In an embodiment, the case **110** may include the first side member **110-1**, the second side member **110-2**, and a support member **110-3** as a third side member. The first side member **110-1** and the second side member **110-2** may be disposed to face each other in a direction substantially perpendicular to a sliding direction of the second structure **140** (e.g., the first direction D**1** and the second direction D**2**). The support member **110-3** may be disposed between the first side member **110-1** and the second side member **110-2** and may be connected to the first side member **110-1** and the second side member **110-2**. The support member **110-3** may connect the first side member **110-1** and the second side member **110-2**. For example, one end portion (e.g., an end portion in the +Y-axis direction as a first end) of the support member **110-3** may be connected with one end portion of the first side member **110-1**, and an opposite end portion (e.g., an end portion in the −Y-axis direction as a second end) of the support member **110-3** may be connected with one end portion of the second side member **110-2**.

(23) In an embodiment, the first side member **110-1**, the second side member **110-2**, and/or the support member **110-3** of the case **110** may be integrally formed. In another embodiment, the first side member **110-1**, the second side member **110-2**, and/or the support member **110-3** may be formed as separate components and may be assembled or fastened together. The first side member **110-1**, the second side member **110-2** and the support member **110-3** may together define an opening of the case **110** at which the second support **140** is slidable into and out of the case **110**. That is, the case **110** may be open in the sliding direction.

(24) In an embodiment, when the front surface (e.g., a surface facing the +Z-axis direction) of the electronic device **100** is viewed (e.g., a view along a third direction, that is, the Z direction), the support member **110-3** may be disposed such that at least a portion of the support member **110-3** overlaps the second structure **140** and/or the display module **150** in the +Z/−Z-axis direction, and only another portion of the support member **110-3** may be exposed in a lateral direction (e.g., the +X/−X-axis direction) of the electronic device **100**. Furthermore, when the rear surface (e.g., a surface facing the −Z-axis direction) of the electronic device **100** is viewed (e.g., a view along the third direction), the support member **110-3** may overlap the back cover **160** in the +Z/−Z-axis direction and may be hidden by the back cover **160**, and thus the support member **110-3** may not be visually exposed to outside the electronic device **100** at the rear thereof.

(25) In an embodiment, the first side member **110-1** and the second side member **110-2** may be disposed on the opposite end portions of the support member **110-3**. For example, the one end portion (e.g., an end portion in the −X-axis direction) of the first side member **110-1** may be connected to the one end portion (e.g., an end portion in the +Y-axis direction) of the support member **110-3**, and the one end portion (e.g., an end portion in the −X-axis direction) of the second side member **110-2** may be connected to the opposite end portion (e.g., an end portion in the +Y-axis direction) of the support member **110-3**. As the second structure **140** and/or the display module **150** slides between the first side member **110-1** and the second side member **110-2**, the second structure **140** and/or at least one portion of the display module **150** may be inserted into the case **110** or may be withdrawn from the case **110**, at through the opening of the case **110**. For example, when the electronic device **100** is changed from the first (closed) state to the second (open) state, the second structure **140** and the at least one portion (e.g., the first region **151**) of the display module **150** may move in the first direction D**1** (e.g., an open direction) between the first side member **110-1** and the second side member **110-2**, and another portion (e.g., a third region **153**) of the display module **150** may move in the second direction D**2**. In contrast, when the electronic device **100** is changed from the second (open) state to the first (closed) state, the second structure **140** and the at least one portion (e.g., the first region **151**) of the display module **150** may move in the second direction D**2** between the first side member **110-1** and the second side member **110-2**, and the other portion (e.g., the third region **153**) of the display module **150** may move in the first direction D**1**.

(26) In an embodiment, the second structure **140** may be configured to slide relative to the case **110**. For example, at least a portion of the second structure **140** may move relative to the case **110** in the first direction D**1** or the second direction D**2** in a state of being substantially parallel to the case **110**.
(27) In an embodiment, the second structure **140** may include the plurality of peripheral portions **143**, **144**, and/or **145** that surround at least a portion of the periphery of the display module **150**. The peripheral portions **143**, **144**, and/or **145** may surround the periphery of the display module **150** at a distal end of the display module **150**. The plurality of peripheral portions **143**, **144**, and/or **145** may include the first peripheral portion **143** extending in a direction perpendicular to the sliding direction of the second structure **140** (e.g., the first direction D**1** or the second direction D**2**), the second peripheral portion **144** that is connected with one end portion (e.g., an end portion in the +Y-axis direction) of the first peripheral portion **143** and that extends in a direction parallel to the sliding direction of the second structure **140** (e.g., the first direction D**1** and the second direction D**2**), and the third peripheral portion **145** that is connected with an opposite end portion (e.g., an end portion in the −Y-axis direction) of the first peripheral portion **143** and that extends in a direction parallel to the sliding direction of the second structure **140** (e.g., the first direction D**1** and the second direction D**2**).
(28) In an embodiment, when the electronic device **100** is in the first state, the first peripheral portion **143** may extend between distal ends of the first side member **110-1** and the second side member **110-2** and may be exposed on the exterior of the electronic device **100** (e.g., exposed to outside the electronic device **100**). At least part of the first peripheral portion **143** may be brought into contact with the distal ends of the first side member **110-1** and the second side member **110-2** in the first state. When the electronic device **100** is changed from the first state to the second state, the first peripheral portion **143** may move in the first direction D**1** and may be spaced apart from the first side member **110-1** and the second side member **110-2**.
(29) In an embodiment, the second peripheral portion **144** and the third peripheral portion **145** may be inserted into (e.g., insertable into) or withdrawn from (e.g., withdrawable from) the first side member **110-1** and the second side member **110-2**, respectively. For example, when the first peripheral portion **143** moves in the first direction D**1**, the second peripheral portion **144** may be withdrawn from the first side member **110-1**, and the third peripheral portion **145** may be withdrawn from the second side member **110-2**. The second and third peripheral portions **144** and **145** may slide together with each other along the sliding direction. In contrast, when the first peripheral portion **143** moves in the second direction D**2**, at least part of the second peripheral portion **144** may be inserted into the first side member **110-1**, and at least part of the third peripheral portion **145** may be inserted into the second side member **110-2**.
(30) In an embodiment, the display module **150** may include the first region **151**, a second region **152**, and the third region **153**. In an embodiment, the second region **152** may extend from the first region **151**, and the third region **153** may extend from the second region **152**. For example, the second region **152** may be located between the first region **151** and the third region **153**. The second region **152** of the display module **150** may be bendable about a bending axis. That is, the first region **151**, the second region **152** and the third region **153** may be in order along the sliding direction. The first to third regions **151** to **153** may together define a planar area of the display module **150**, without being limited thereto.
(31) In an embodiment, the display module **150** may be disposed on the second structure **140**. The display module **150** may be disposed on one surface of the second structure **140** such that at least a portion of the periphery of the display module **150** is surrounded by the plurality of peripheral portions **143**, **144**, and/or **145** of the second structure **140**. For example, the display module **150** may be disposed on the second structure **140** such that at least a partial region of the display module **150** faces a front direction of the electronic device **100** (e.g., the +Z-axis direction). The display module **150** may be configured to move together with the second structure **140** when the

second structure **140** slides relative to the case **110**. For example, the display module **150** may be attached to the second structure **140** by an adhesive material (e.g., a double-sided tape or glue).

(32) According to an embodiment, as the electronic device **100** is changed between the first state and the second state, an exposed region of the display module **150** exposed in the front direction of the electronic device **100** may be expanded or reduced. A planar area of the display module **150** which is disposed at the front of the electronic device **100** is changeable (e.g., variable), depending upon the open or closed state of the electronic device **100**. For example, in the first state, the first region **151** may form a front display region visually exposed on the front surface of the electronic device **100**. In the first state, only the first region **151** may define the front display region (e.g., display surface). In the second state, at least a portion of the second region **152**, together with the first region **151**, may form the front display region visually exposed on the front surface of the electronic device **100**. The front display region may be defined as a region in which a predetermined screen is provided on the front surface of the electronic device **100** (e.g., a display surface). A rear display region may be defined as a region in which a predetermined screen is provided on the rear surface of the electronic device **100** (e.g., a display surface).

(33) In an embodiment, the display module **150** may be formed of a flexible material such that, depending on operational states (e.g., the first state and the second state) of the electronic device **100**, at least one portion of the display module **150** faces the front direction of the electronic device **100** and another portion of the display module **150** faces a rear direction of the electronic device **100**.

(34) In an embodiment, the first region **151** may form a portion of the front surface of the electronic device **100** in both the first state and the second state. In the first state, an entirety of the front surface is the first region **151** (e.g., only the first region **151**), while in the second state, the front surface includes an entirety of the first region **151** together with a portion (or an entirety) of the second region **152**. In an embodiment, the second region **152** may be visually exposed at the rear surface of the electronic device **100** (e.g., a window region **162**) in the first state. The second region **152** may form the front surface of the electronic device **100** together with the first region **151**, in the second state. In an embodiment, the third region **153** may not be exposed by being hidden by the back cover **160** in the first state and may be visually exposed to outside the rear surface of the electronic device **100** (e.g., the window region **162**), in the second state.

(35) In an embodiment, the back cover **160** may form at least a portion of the rear surface of the electronic device **100**. For example, the back cover **160** may be disposed between the first side member **110-1** and the second side member **110-2**. In this case, the back cover **160** may be disposed to at least partially overlap the support member **110-3** such that the support member **110-3** is not exposed on the rear surface of the electronic device **100**.

(36) In an embodiment, the back cover **160** may include the window region **162** and an opaque region **163**. The window regions **162** may be formed of a transparent or translucent material. For example, at least a partial region of the back cover **160** may be formed of (or include) a transparent or translucent material, and thus the window region **162** may be formed (or provided). At least a portion (e.g., the opaque region **163**) of the back cover **160** may be formed of a polymer, coated or colored glass, ceramic, metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the aforementioned materials.

(37) In an embodiment, the second region **152** or the third region **153** of the display module **150** may be visually exposed at the rear surface of the electronic device **100**, through the window region **162**. For example, in the first state of the electronic device **100**, at least a portion of the second region **152** may be visually exposed in the rear direction of the electronic device **100** through the window region **162** (refer to FIG. 1B). In the second state of the electronic device **100**, at least a portion of the third region **153** may be visually exposed in the rear direction of the electronic device **100** through the window region **162** (refer to FIG. 2B).

(38) In an embodiment, the electronic device **100** may further include an audio module (e.g., **102**

and **103**). The audio module (e.g., **102** and **103**) may include the microphone hole **102** and the speaker hole **103**. A microphone for obtaining external sound may be disposed in the microphone hole **102**. The speaker hole **103** may include an external speaker hole **103** and/or a receiver hole (not illustrated) for telephone call. In another embodiment, the speaker hole **103** and the microphone hole **102** may be implemented with a single hole, or a speaker (e.g., a piezoelectric speaker) may be included without the speaker hole **103**.

(39) In an embodiment, the electronic device **100** may further include a front camera module (not illustrated) that is visually exposed on the front surface of the electronic device **100** and a rear camera module **108** that is visually exposed on the rear surface of the electronic device **100**. Although not illustrated, the front camera module (not illustrated) may be visually exposed through at least a partial region of the display module **150**. In another embodiment, the front camera module (e.g., an under display camera (UDC)) may be disposed under the display module **150**. For example, the front camera module (not illustrated) may be at least partially disposed under the display module **150** and may be configured to take an image of an object through a portion of an active region of the display module **150**. In the other embodiment, the front camera module (not illustrated) may not be visually exposed on a region of the display module **150**. The rear camera module **108** may include a plurality of camera modules (e.g., a dual camera or a triple camera). However, the rear camera module **108** is not necessarily limited to including the plurality of camera modules and may be implemented with one camera module in some embodiments. The front camera module (not illustrated) and the rear camera module **108** may include one or more lenses, an image sensor, and/or an image signal processor. Although not illustrated, in another embodiment, the rear camera module **108** and a flash (not illustrated) may be disposed on the rear surface of the electronic device **100**. The flash (not illustrated) may include, for example, a light emitting diode or a xenon lamp. In some embodiments, two or more lenses (e.g., a wide angle lens and/or a telephoto lens) and image sensors may be disposed on one surface of the electronic device **100**.

(40) In an embodiment, the electronic device **100** may further include a connector hole **106**. The connector hole **106** may include a first connector hole **106** capable of accommodating a connector (e.g., a USB connector) for transmitting and receiving power and/or data with an external electronic device and/or a second connector hole (not illustrated) (e.g., an earphone jack) capable of accommodating a connector for transmitting and receiving audio signals with an external electronic device.

(41) In another embodiment (not illustrated), the electronic device **100** may further include a key input device (not illustrated) (e.g., an input device **350** of an electronic device **301** of FIG. **11**). The key input device (not illustrated) may be disposed on the side surface of the electronic device **100**. For example, the key input device (not illustrated) may be formed in a button type and may be disposed on the first side member **110-1** and/or the second side member **110-2**. Furthermore, for example, the key input device (not illustrated) may be implemented in a form such as a soft key on the display module **150**.

(42) In another embodiment (not illustrated), the electronic device **100** may further include a sensor module (not illustrated). The sensor module (not illustrated) (e.g., a sensor module **376** of the electronic device **301** of FIG. **11**) may generate an electrical signal or a data value that corresponds to an operational state inside the electronic device **100** or an environmental state external to the electronic device **100**. For example, the sensor module may include at least one of a proximity sensor, an HRM sensor, a fingerprint sensor, a time of flight (TOF) sensor, an ultrasonic sensor, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biosensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(43) As described above, the electronic device **100** according to the embodiment may be changeable between the first state (e.g., the state of FIG. **1**) and the second state (e.g., the state of

FIG. **2**), as the second structure **140** slides (together with the display module **150**) relative to the case **110**. For example, the first state may refer to a closed state, and the second state may refer to an opened state.

(44) Referring to FIG. **1**, the first state may be defined as a state in which the first region **151** of the display module **150** forms an entirety of the front surface of the electronic device **100**, and a portion of the second region **152** of the display module **150** is visually exposed to outside the back cover **160** at the rear surface of the electronic device **100** through the window region **162** of the back cover **160**. For example, in the first state, the first region **151** may form the front display region (e.g., a region in which a display screen is provided at the front surface of the electronic device **100** as a display region or a display area). In this case, a portion of the second region **152** may form the rear display region (e.g., a region in which a display screen is provided at the rear surface of the electronic device **100** as a display region or a display area). In an embodiment, the rear display region may be configured to receive a touch input from the outside through the window region **162** of the back cover **160** in the first state (e.g., a touch area).

(45) Referring to FIG. **2**, the second state may be defined as a state in which at least a portion of the second region **152** of the display module **150** forms the front surface of the electronic device **100** together with the first region **151**. For example, as at least a portion of the second region **152**, together with the first region **151**, forms the front display region in the second state, the front display region may be expanded, compared to that in the first state. That is, the electronic device **100** which is open and which is closed, may respectively include a planar area of the display screen at the front surface of the open electronic device **100**, where the planar area of the open electronic device is larger than the planar area of the closed electronic device. In this case, the third region **153** of the display module **150** may be visually exposed at the rear surface of the electronic device **100** in the second state, through the window region **162**. The third region **153** may prevent parts disposed in the electronic device **100** from being visually exposed on the rear surface of the electronic device **100** through the window region **162**. In an embodiment, the third region **153** may be configured so as not to form the rear display region (e.g., a non-display region or non-display area) or receive a touch input through the window region **162** of the back cover **160** (e.g., a non-touch area). However, the disclosure is not limited thereto, and in another embodiment, the third region **153** may be configured to form the rear display region or receive a touch input through the window region **112** in the second state.

(46) FIG. **3** is an exploded perspective view of the electronic device according to an embodiment.

(47) Referring to FIG. **3**, the electronic device **100** according to an embodiment may include the case **110**, a first structure **120**, a guide member **130**, the second structure **140**, the display module **150**, the back cover **160**, and a circuit board **190**.

(48) In an embodiment, the case **110** may include the support member **110-3** disposed on an upper surface (e.g., a surface facing the +Z-axis direction) of the back cover **160**, and the first side member **110-1** and the second side member **110-2** disposed on the opposite end portions of the support member **110-3** in a lengthwise direction (e.g., the Y-axis direction) to face each other. The support member **110-3**, the first side member **110-1**, and the second side member **110-2** may be combined together to form a space in which at least a part of other components (e.g., the circuit board **190**, the first structure **120**, or the second structure **140**) of the electronic device **100** is disposed.

(49) In an embodiment, the support member **110-3** may include an opening region **111** penetrating at least a partial region of the support member **110-3** in an up/down direction (e.g., the Z-axis direction). The support member **110-3** may have a frame shape, owing to the opening region **111**. The circuit board **190** and the back cover **160** may be disposed to at least partially face each other through the opening region **111**. For example, the support member **110-3** may be disposed between the circuit board **190** and the back cover **160**, and a lower surface (e.g., a surface facing the −Z-axis direction) of the circuit board **190** and the upper surface (e.g., a surface facing the +Z-axis

direction) of the back cover **160** may face each other through the opening region **111**. As the opening region **111** is formed in the support member **110-3**, the circuit board **190** and a partial region of the back cover **160** (e.g., a conductive region **165** of FIG. **8**) may be electrically connected as will be described below with reference to FIG. **6**.

(50) In an embodiment, the first side member **110-1** and the second side member **110-2** may include recesses **113** and **115** formed in partial regions of peripheral portions thereof, respectively, in a direction substantially parallel to the sliding direction of the second structure **140** (e.g., the +X-axis direction or the −X-axis direction). The recesses **113** and **115** may provide spaces that at least parts of the second peripheral portion **144** and the third peripheral portion **145** of the second structure **140** are inserted into or withdrawn from when the second structure **140** slides relative to the first structure **120**. For example, when the electronic device **100** is changed to the first state (e.g., the state of FIG. **1**) or the second state (e.g., the state of FIG. **2**), the second peripheral portion **144** may move through the recess **113** of the first side member **110-1**, and the third peripheral portion **145** may move through the recess **115** of the second side member **110-2**.

(51) In an embodiment, the first structure **120** may be fixed and/or coupled to the case **110**. For example, the first structure **120** and the case **110** may be the basis for sliding of the second structure **140**. According to the illustrated embodiment, the first structure **120** and the case **110** may be formed as separate parts and may be assembled and/or coupled with each other. However, the disclosure is not limited thereto, and according to an embodiment to which the disclosure is applied, the first structure **120** and the case **110** may be integrally formed with each other and may be implemented as one part.

(52) In an embodiment, the first structure **120** may include a first surface **121** (e.g., an upper surface or a surface facing the +Z-axis direction), a second surface **122** (e.g., a lower surface or a surface facing the −Z-axis direction) that faces away from the first surface **121**, and a plurality of side surfaces **123**, **124**, **125**, and **126** which provide a space together with the first surface **121** and the second surface **122**. The plurality of side surfaces **123**, **124**, **125**, and **126** may include the first side surface **123** extending in a direction (e.g., the Y-axis direction) perpendicular to the sliding direction of the second structure **140**, the second side surface **124** facing the first side surface **123**, and the third side surface **125** and the fourth side surface **126** connecting the first side surface **123** and the second side surface **124** and facing each other. For example, the third side surface **125** may face the first side member **110-1**, and the fourth side surface **126** may face the second side member **110-2**.

(53) In an embodiment, the first structure **120** may be formed of (or include) a polymer resin (e.g., polypropylene, polyethylene, polystyrene, polyethylene terephthalate, polyamide, polyester, polyvinyl chloride, polyurethane, polycarbonate, or polyvinylidene chloride) or metal.

(54) In an embodiment, at least a portion of the first structure **120** may be surrounded by the second structure **140**. For example, the first surface **121**, the second side surface **124**, and at least a portion of the second surface **122** of the first structure **120** may be covered by the second structure **140**. The first structure **120** may be connected with the second structure **140** such that the second structure **140** slides relative to the first structure **120**.

(55) In an embodiment, the guide member **130** may connect and/or support the first structure **120** and at least a portion of the second structure **140** such that the second structure **140** is able to slide (e.g., is slidable). The guide member **130** may include a first roller member **132** (e.g., a roller), at least one second roller member **134**, and at least one belt member **136** (e.g., a belt).

(56) In an embodiment, the first roller member **132** may be disposed on the second side surface **124** of the first structure **120**, and the second roller member **134** may be disposed on the first side surface **123** of the first structure **120**. For example, the first roller member **132** may be disposed to face toward the second side surface **124** from outside the first structure **120** and may be coupled so as to be rotatable relative to the first structure **120**. The second roller member **134** may be disposed to face toward the first side surface **123** from inside the first structure **120** and may be coupled so

as to be rotatable relative to the first structure **120**. In another embodiment, the first roller member **132** may include a plurality of rollers.
(57) In an embodiment, the belt member **136** may be disposed inside the first structure **120** to at least partially surround the second roller member **134**. For example, at least one portion of the belt member **136** may be accommodated in the first structure **120** (e.g., at a rear side), and another portion of the belt member **136** may be exposed outside the first structure **120** (e.g., at a front side). Opposite end portions of the belt member **136** may be connected to different portions of the second structure **140**, respectively. The belt member **136** may be implemented with one strap (or, band), or may be implemented in a form in which two or more straps (or, bands) are fastened with each other. In an embodiment, the belt member **136** may be formed of metal and/or a polymer resin.
(58) A coupling relationship between the first roller member **132**, the second roller member **134**, the belt member **136**, and the second structure **140** will be described below in detail with reference to FIG. **5**.
(59) In an embodiment, the first structure **120** may have, on the first surface **121**, a depression **128** in which a portion of the belt member **136** exposed outside the first structure **120** is disposed. For example, a partial region of the first surface **121** of the first structure **120** may be recessed toward the second surface **122** to form the depression **128**. At least a portion of the belt member **136** may be disposed in the depression **128** and may move together along the depression **128**, in the sliding direction of the second structure **140** (e.g., the X-axis direction) when the second structure **140** slides. According to the illustrated embodiment, the belt member **136** may include a plurality of belts, and a plurality of depressions **128** corresponding to the plurality of belts may be formed on the first structure **120**. However, the disclosure is not necessarily limited to the illustrated embodiment, and according to an embodiment to which the disclosure is applied, the numbers and/or positions of belt members **136** and depressions **128** may be diversely modified. Furthermore, the belt member **136** and the depression **128** may have different lengths depending on the position in which the belt member **136** is disposed.
(60) In an embodiment, the second structure **140** may be connected with the first structure **120** so as to be slidable relative to the first structure **120**. For example, the second structure **140** may slide relative to the first structure **120** and the case **110**, and may move in the +X/−X-axis direction relative to the first structure **120** and the case **110** that are relatively fixed in position.
(61) In an embodiment, the second structure **140** may be disposed to surround at least a portion of the first structure **120**. For example, the second structure **140** may surround the first surface **121**, the second side surface **124**, and a partial region of the second surface **122** of the first structure **120**. The first surface **121** and the second side surface **124** of the first structure **120** may be covered by the second structure **140** irrespective of operational states (e.g., the first state and the second state) of the electronic device **100**, and a region where the second surface **122** of the first structure **120** is covered by the second structure **140** may be expanded or reduced depending on operational states of the electronic device **100**.
(62) In an embodiment, the second structure **140** may support the display module **150**. For example, the second structure **140** may be closely fixed to the display module **150**. The display module **150** which is fixed to the second structure **140**, is movable together with the second structure **140**. At least a portion of the second structure **140** may be attached to the display module **150** through an adhesive member (e.g., a double-sided tape or glue) disposed between the display module **150** and the second structure **140**. The second structure **140**, together with the display module **150**, may slide relative to the first structure **120**.
(63) In an embodiment, the second structure **140** may include a first support portion **141** and a second support portion **142**. The second support portion **142** may extend from the first support portion **141**. In an embodiment, the first support portion **141** may support at least one portion of the first region **151** of the display module **150**, and the second support portion **142** may support another portion of the first region **151**, the second region **152**, and/or the third region **153** of the display

module **150**. For example, the first support portion **141** may be disposed to face the one portion of the first region **151**. The second support portion **142** may be disposed to face the other portion of the first region **151**, the second region **152**, and/or the third region **153**.
(64) In an embodiment, the first support portion **141** may include the first peripheral portion **143**, the second peripheral portion **144**, and the third peripheral portion **145**. The first peripheral portion **143**, the second peripheral portion **144**, and the third peripheral portion **145** may surround at least a portion of the first region **151** of the display module **150**. The first support portion **141** may be formed of a substantially flat plate. For example, when the second structure **140** slides, the first support portion **141** may move in a state of being substantially parallel to the first structure **120** without being deformed.
(65) In an embodiment, the second support portion **142** may be formed of a bendable material so as to at least partially form a curved surface when the second structure **140** slides. For example, a bending portion of the second support portion **142** may vary depending on operational states (e.g., the first state and the second state) of the electronic device **100**. That is, the bending portion may be disposed flat or bent, depending upon the state of the electronic device **100**. The second support portion **142** may support the second region **152** such that the second region **152** of the display module **150** slides while forming a curved surface in at least a partial region.
(66) In an embodiment, the second support portion **142** may include a plurality of protrusions **147** that form a multi-joint structure. The protrusions **147** may protrude toward the first structure **120** in a state in which the second structure **140** surrounds the first structure **120**. For example, in a state in which at least some of the protrusions **147** make contact with the first roller member **132**, the protrusions **147** may rotate and/or linearly move as the second structure **140** slides.
(67) In an embodiment, the protrusions **147** may have a predetermined length in a direction (e.g., the Y-axis direction) perpendicular to the sliding direction of the second structure **140**. The protrusions **147** may be spaced apart from each other by a predetermined gap in a direction (e.g., the X-axis direction) substantially parallel to the sliding direction of the second structure **140**. Meanwhile, although not illustrated, according to another embodiment, the second support portion **142** may include a flexible film (not illustrated) that extends from and/or connects to one side of the first support portion **141**, and the protrusions **147** may be disposed on one surface of the flexible film. In this case, the display module **150** may be disposed on an opposite surface of the flexible film that faces away from the one surface of the flexible film.
(68) In an embodiment, the display module **150** may be formed to be flexible so as to at least partially form a curved surface. For example, the display module **150** may be formed such that at least a partial region is bent and one part at least partially faces another part accordingly. The display module **150** may include, for example, a flexible display or a foldable display. According to an embodiment to which the disclosure is applied, the entire region of the display module **150** may be formed of a flexible material. Alternatively, one partial region of the display module **150** may be formed of a flexible material, and another partial region of the display module **150** may be formed of a non-flexible material that is not bent.
(69) In an embodiment, the display module **150** may include the first region **151**, the second region **152**, and the third region **153**. The second region **152** may extend from the first region **151**, and the third region **153** may extend from the second region **152**. A display region of the display module **150** on which a predetermined screen is displayed may be changed based on an area visually exposed on the front surface and/or the rear surface of the electronic device **100**. For example, in the first state, the first region **151** may be visually exposed on the front surface of the electronic device **100**, and the second region **152** may be visually exposed on the rear surface of the electronic device **100**. In the second state, the first region **151** and the second region **152** may be visually exposed on the front surface of the electronic device **100**.
(70) In an embodiment, the display module **150** may be disposed on the second structure **140** and may move together with the second structure **140**, relative to the first structure **120** and the case

110. For example, the first region **151** may be partially fixed to the first support portion **141** and the second support portion **142**, and the second region **152** and the third region **153** may be fixed to the second support portion **142**. The positions and/or deformation of the regions (e.g., the first region **151**, the second region **152**, and the third region **153**) of the display module **150** depending on operational states (e.g., the first state and the second state) of the electronic device **100** will be described below in more detail with reference to FIG. **4**.

(71) In an embodiment, at least a portion of the periphery of the third region **153** of the display module **150** or a rear surface of the third region **153** may be a portion electrically connected to the circuit board **190**. For example, at least a portion of the periphery of the third region **153** of the display module **150** may extend in one direction (e.g., the +X-axis direction) and may be electrically connected to the circuit board **190**. In another example, the display module **150** may include a connector **155** disposed on the rear surface of the third region **153** of the display module **150** and may be electrically connected to the circuit board **190** through a flexible printed circuit board (FPCB) (not illustrated) that is electrically connected to the connector **155**.

(72) In an embodiment, a driver (not illustrated) for driving light emitting elements (e.g., OLEDs) included in the display module **150** may be disposed on the rear surface of the third region **153** of the display module **150** or an extension of the third region **153**. The driver may include a drive circuit and may have a chip on film (COF) structure. As a part is disposed on the rear surface of the third region **153** of the display module **150** or the extension of the third region **153**, the distance from the circuit board **190** may be decreased, and thus electrical noise may be reduced.

(73) In an embodiment, the back cover **160** may be disposed under the support member **110-3** (e.g., in the −Z-axis direction). The back cover **160** may include the window region **162** formed of a transparent or translucent material and the opaque region **163** surrounding the periphery of the window region **162**. For example, when the rear surface of the electronic device **100** is viewed in the first state (e.g., the state of FIG. **1**), the window region **162** may allow the second region **152** of the display module **150** to be visually exposed on the rear surface (or, the back cover **160**) of the electronic device **100**.

(74) In an embodiment, the circuit board **190** may be disposed between the first structure **120** and the back cover **160**. For example, an upper surface of the circuit board **190** may face the second surface **122** of the first structure **120**, and the lower surface of the circuit board **190** may face the opening region **111** of the support member **110-3** and/or the back cover **160**. Here, the upper surface of the circuit board **190** may refer to a surface substantially facing the +Z-axis direction or a surface facing toward the first structure **120**, and the lower surface may refer to a surface facing away from the upper surface.

(75) In an embodiment, various electronic parts included in the electronic device **100** may be electrically connected to the circuit board **190**. The circuit board **190** may include a printed circuit board (PCB) and/or a flexible printed circuit board (FPCB).

(76) In an embodiment, a processor (e.g., a processor **320** of FIG. **11**), a memory (e.g., a memory **330** of FIG. **11**), and/or an interface (e.g., an interface **377** of FIG. **11**) may be mounted on the circuit board **190**. For example, the processor (e.g., the processor **32** of FIG. **11**) may include a main processor (e.g., a main processor **321** of FIG. **11**) and/or an auxiliary processor (e.g., an auxiliary processor **323** of FIG. **11**), and the main processor (e.g., the main processor **321** of FIG. **11**) and/or the auxiliary processor (e.g., the auxiliary processor **323** of FIG. **11**) may include one or more of a central processing unit, an application processor, a graphic processing unit, an image signal processor, a sensor hub processor, or a communication processor. For example, the memory may include a volatile memory or a nonvolatile memory. For example, the interface may include a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. Furthermore, the interface may electrically or physically connect the electronic device **100** with an external electronic device and may include a USB connector, an SD card/MMC connector, or an audio connector.

(77) FIGS. **4**A and **4**B (otherwise referred to as FIG. **4**) are sectional views of the electronic device according to an embodiment.
(78) FIG. **4**A is a sectional view illustrating the first state of the electronic device, and FIG. **4**B is a sectional view illustrating the second state of the electronic device. FIG. **4** illustrates a section taken along line A-A′ of FIG. **3**.
(79) Referring to FIG. **4**, the electronic device **100** according to an embodiment may include the case **110**, the first structure **120**, the second structure **140**, the display module **150**, and the back cover **160**. Meanwhile, at least some of the components of the electronic device **100** illustrated in FIG. **4** are identical or similar to the components of the electronic device **100** described with reference to FIGS. **1** to **3**, and therefore repetitive descriptions will hereinafter be omitted.
(80) According to an embodiment, the first structure **120** may include the first roller member **132**, the second roller member **134**, and the belt member **136**. The second structure **140** may include the first support portion **141** and the second support portion **142** extending from the first support portion **141** and may be slidably connected to the first structure **120** so as to slide. The display module **150** may include the first region **151**, the second region **152** extending from the first region **151**, and the third region **153** extending from the second region **152** and may be disposed on at least one surface of the second structure **140**. The back cover **160** may include the window region **162** and the opaque region **163** extending from the window region **162**.
(81) In an embodiment, the second structure **140** may be disposed to surround at least a portion of the first structure **120**. For example, at least one portion of the second structure **140** may be disposed over the first structure **120** (e.g., in the +Z-axis direction), and another portion of the second structure **140** may be disposed under the first structure **120** (e.g., in the −Z-axis direction).
(82) In an embodiment, the first support portion **141** of the second structure **140** may be disposed over the first structure **120** (e.g., in the +Z-axis direction), and at least part of the second support portion **142** may be disposed under the first structure **120** (e.g., in the −Z-axis direction).
(83) In an embodiment, at least part of the second support portion **142** may be disposed in a space between the first structure **120** and the back cover **160**. The area of the second support portion **142** disposed between the first structure **120** and the back cover **160** may vary depending on operational states (e.g., the first state and the second state) of the electronic device **100**. For example, as the second structure **140** slides, at least part of the second support portion **142** may be inserted into or withdrawn from the space between the first structure **120** and the back cover **160**. In an embodiment, the display module **150** may move together with the second structure **140**. For example, as the second structure **140** slides, at least a portion of the second region **152** of the display module **150**, together with the second support portion **142**, may be withdrawn from or inserted into the space between the first structure **120** and the back cover **160**. As the second region **152** is withdrawn and/or inserted by the sliding of the second structure **140**, a portion of the second region **152** where a curved surface is formed may move. For example, in the first state, the curved surface may be formed on one end portion of the second region **152** connected with the first region **151** (e.g., FIG. **4**A), and in the second state, the curved surface may be formed on an opposite end portion of the second region **152** connected with the third region **153** (e.g., FIG. **4**B). The curved surface may move in the direction from the one end portion to the opposite end portion of the second region **152** in a process in which the electronic device **100** is changed from the first state to the second state.
(84) In an embodiment, when at least a portion of the second region **152** is withdrawn from between the first structure **120** and the back cover **160**, the display region visually exposed on the front surface (e.g., a surface facing the +Z-axis direction) of the electronic device **100** may be expanded. In contrast, when the at least a portion of the second region **152** is inserted between the first structure **120** and the back cover **160**, the display region visually exposed on the front surface of the electronic device **100** may be reduced.
(85) In an embodiment, the second structure **140** may be slidably connected to the first structure

120 by the first roller member **132**, the second roller member **134**, and the belt member **136**. Referring to FIG. **3**, a portion of the second roller member **134** may protrude outside the first structure **120**, so as to be exposed at a front side of the first structure **120**.
(86) In an embodiment, the second support portion **142** may be disposed to surround (or extend along) at least a portion of the first roller member **132**, and the belt member **136** may be disposed to surround (or extend along) at least a portion of the second roller member **134**. The first roller member **132** and the second roller member **134** may be disposed so as to be rotatable relative to the first structure **120** (e.g., the first structure **120** of FIG. **3**). The rollers may rotate about a rotation axis extended along the Y-axis direction. In this case, the opposite end portions of the belt member **136** may be respectively connected to the first support portion **141** and the second support portion **142** of the second structure **140**. Opposing ends of the belt member **136** may be spaced apart from each other along the second structure **140**. For example, as the second structure **140** and the belt member **136** are connected with each other at two locations, and the first roller member **132** and the second roller member **134** are disposed between the second structure **140** and the belt member **136**, the second structure **140** and the belt member **136** may be slid by rotation of the first roller member **132** and the second roller member **134**. In an embodiment, other components of the electronic device **100** (e.g., the circuit board **190** of FIG. **3**) may be disposed in a space between the belt member **136** and the second structure **140**.
(87) In an embodiment, the electronic device **100** may include the first state (e.g., the state of FIG. **1**) and the second state (e.g., the state of FIG. **2**). For example, the electronic device **100** may be changed to the first state or the second state as the second structure **140** and the display module **150** move in the first direction D**1** or the second direction D**2** relative to the case **110**, the first structure **120**, and the back cover **160**. In this case, the first state and the second state of the electronic device **100** may be determined depending on the positions of the second structure **140** and the display module **150**, relative to other components of the electronic device **100**.
(88) In an embodiment, when the electronic device **100** is in the first state, the electronic device **100** may be changed to the second state by sliding at least a portion (e.g., the first support portion **141**) of the second structure **140** in the first direction D**1**. In contrast, when the electronic device **100** is in the second state, the electronic device **100** may be changed to the first state by sliding at least a portion (e.g., the first support portion **141**) of the second structure **140** in the second direction D**2**.
(89) In an embodiment, the first peripheral portion **143** of the second structure **140** may be brought into contact with or spaced apart from the second side member **110-2** of the case **110**, as the second structure **140** slides. For example, in the first state, the first peripheral portion **143** may make contact with the second side member **110-2** to form substantially the same plane (e.g., a single planar surface, refer to FIG. **1**), and in the second state, the first peripheral portion **143** may be spaced apart from the second side member **110-2** in the first direction D**1** (e.g., refer to FIG. **2**). In an embodiment, the second side member **110-2** may further protrude in the +Z-axis direction by a specified height beyond a partial region of the display module **150** that faces toward the front surface of the electronic device **100**. For example, in the first state, the second side member **110-2** may be formed to be higher than a portion of the first region **151** in the +Z-axis direction, and in the second state, the second side member **110-2** may be formed to be higher than the first region **151** and a portion of the second region **152** in the +Z-axis direction. Although FIG. **4** illustrates the section taken along line A-A′ of FIG. **3** and therefore the first side member (e.g., the first side member **110-1** of FIGS. **1** to **3**) is not illustrated in FIG. **4**, the position between the second side member **110-2** and the first peripheral portion **143** and a different in height between the second side member **110-2** and the display module **150** may be identically applied to the first side member **110-1**.
(90) Referring to FIG. **4**A, in an embodiment, when the electronic device **100** is in the first state, the first region **151** of the display module **150** may form a display region visually exposed on the

front surface of the electronic device **100**, and at least a portion of the second region **152** may be disposed between the first structure **120** and the back cover **160** and may face the rear direction of the electronic device **100** (e.g., the +Z-axis direction). In this case, at least a portion of the second regions **152** may be disposed to face the window region **162** of the back cover **160**, and at least a portion of the third region **153** may be disposed to face the opaque region **163** of the back cover **160**. For example, when the electronic device **100** is viewed in the rear direction, at least a portion of the second region **152** may be visually exposed on the rear surface of the electronic device **100** through the window region **162**, and the third region **153** may be hidden by the opaque region **163** and may not be visually exposed.

(91) According to an embodiment, in the first state, the first region **151** of the display module **150** may form a display region visually exposed on the front surface of the electronic device **100**, and the second region **152** may form a display region visually exposed on the rear surface of the electronic device **100**. For example, the second region **152** visually exposed on the rear surface of the electronic device **100** through the window region **162** of the back cover **160** in the first state may be configured to receive a touch input from the outside, or may be configured to display a predetermined screen (e.g., provide a display screen at which an image is displayed). Portions of the first region **151** and the second region **152** may face outward laterally at a side of the electronic device **100**, so as to form a display region visually exposed.

(92) Referring to FIG. **4**B, in an embodiment, when the electronic device **100** is in the second state, the first region **151** and the second region **152** of the display module **150** may form a display region visually exposed on the front surface of the electronic device **100**. For example, when the electronic device **100** is changed from the first state to the second state, at least a portion of the second region **152** facing the back cover **160** may be withdrawn from between the first structure **120** and the back cover **160** and/or between the first roller member **132** and the back cover **160**, and may move along the lateral side of the electronic device **100** to the front side of the electronic device **100**. As the electronic device **100** is changed to the second state, a portion of the second region **152** that is disposed inside the case **110** of the electronic device **100** in the first state and is not exposed, may be visually exposed on the front surface of the electronic device **100**, and thus a planar area of the display region visually exposed on the front surface of the electronic device **100** may be expanded. In this case, as the second region **152** moves, at least a portion of the third region **153** may be disposed to face the window region **162**. The third region **153** may prevent other components disposed in the electronic device **100** from being visually exposed through the window region **162** when the electronic device **100** is in the second state.

(93) In another embodiment, the third region **153** of the display module **150** may be configured to be different from the first region **151** and/or the second region **152**. For example, the first region **151** and the second region **152** may be configured to display a predetermined screen (e.g., provide a display screen, a display region and/or a display area), or receive a touch input (e.g., provide a touch area), in the first state or the second state. Unlike the first region **151** and/or the second region **152**, the third region **153** may serve to prevent components at the inside of the electronic device **100** from being visually exposed through the window region **162**. The third region **153** may not include a component (e.g., a display panel **230**) for displaying a screen and/or a component (e.g., a touch panel **220**) for a touch input. However, this is illustrative, and according to an embodiment to which the disclosure is applied, the third region **153** may be configured to display a screen or receive a touch input.

(94) FIG. **5** is a sectional view illustrating a coupling relationship between the second structure, the first roller member, the second roller member, and the belt member of the electronic device according to an embodiment.

(95) FIG. **5** may be a sectional view in which some of the components in FIG. **4** (e.g., the case, the first structure, and/or the back cover) are omitted for convenience of description, and repetitive descriptions will hereinafter be omitted.

(96) Referring to FIG. **5**, the electronic device according to an embodiment (e.g., the electronic device **100** of FIG. **4**) may include the first roller member **132**, the second roller member **134**, the belt member **136**, the second structure **140**, and the display module **150**.
(97) In an embodiment, the opposite end portions **136***a* and **136***b* of the belt member **136** may be respectively connected to the first support portion **141** and the second support portion **142** of the second structure **140**, such as being fixedly connected thereto. For example, the one end portion **136***a* (e.g., a first end portion) of the belt member **136** may be connected to a coupling protrusion **148** of the first support portion **141**. The opposite end portion **136***b* (e.g., a second end portion) of the belt member **136** may be connected to at least part of the second support portion **142**. The opposite end portion **136***b* of the belt member **136** may be fixedly disposed between the second support portion **142** and the display module **150**.
(98) In an embodiment, the belt member **136** may be connected to the first support portion **141** and the second support portion **142** and may provide tension to pull one end of the second support portion **142** with respect to the first support portion **141**. For example, the belt member **136** may be disposed between the first support portion **141** and the second support portion **142** and may allow the second support portion **142** to remain in a tight state, by pulling a second edge **142***b* of the second support portion **142** in a direction (e.g., the +X-axis direction) opposite to a direction toward a first edge **142***a* of the second support portion **142**. Accordingly, the second support portion **142** formed to be bendable may remain in the tight state and may stably support the display module **150**.
(99) As the first support portion **141** and the second support portion **142** are connected to each other by the belt member **136**, the first edge **142***a* and the second edge **142***b* of the second support portion **142** may move along the sliding direction, while maintaining a predetermined gap when the second structure **140** slides. For example, when the first edge **142***a* moves in the first direction D**1** as the second structure **140** slides, the second edge **142***b* may move in the second direction D**2**. In contrast, when the first edge **142***a* moves in the second direction D**2**, the second edge **142***b* may move in the first direction D**1**.
(100) In an embodiment, the first roller member **132** and the second roller member **134** may form a first rotational axis and a second rotational axis that extend in a direction substantially perpendicular to the sliding direction of the second structure **140** (e.g., the first direction D**1** or the second direction D**2**). The first roller member **132** and the second roller member **134** may rotate about the respective rotational axes (e.g., the first rotational axis and the second rotational axis) in the clockwise or counterclockwise direction depending on the sliding direction of the second structure **140** (e.g., the first direction D**1** or the second direction D**2**) when the second structure **140** slides. For example, when the first support portion **141** moves in the first direction D**1**, the first roller member **132** and the second roller member **134** may rotate in the counterclockwise direction. In contrast, when the first support portion **141** moves in the second direction D**2**, the first roller member **132** and the second roller member **134** may rotate in the clockwise direction. When the first support portion **141** moves, the first roller member **132** and the second roller member **134** may rotate together, and the second support portion **142** and the belt member **136** may move.
(101) In an embodiment, the opposite end portions **136***a* and **136***b* of the belt member **136** may be moved in opposite directions by the first roller member **132** and the second roller member **134** when the second structure **140** slides. For example, when the first support portion **141** moves in the first direction D**1**, the one end portion **136***a* of the belt member **136** may move in the first direction D**1**, and the opposite end portion **136***b* of the belt member **136** may move in the second direction D**2**. In contrast, when the first support portion **141** moves in the second direction D**2**, the one end portion **136***a* of the belt member **136** may move in the second direction D**2**, and the opposite end portion **136***b* of the belt member **136** may move in the first direction D**1**.
(102) FIGS. **6**A and **6**B (otherwise referred to as FIG. **6**) illustrate an operation in which a touch is input through the back cover in the first state of the electronic device according to an embodiment.

FIG. **6**B is an enlarged view of region A shown in FIG. **6**A. FIGS. **7**A, **7**B and **7**C (otherwise referred to as FIG. **7**) illustrate the back cover of the electronic device according to an embodiment. FIGS. **7**B and **7**C illustrate a section taken along line B-B′ of FIG. **7**A.

(103) Referring to FIGS. **6** and **7**, the electronic device **100** according to an embodiment may include the first structure **120**, the second structure **140**, the display module **150**, and the back cover **160**.

(104) In an embodiment, the second structure **140** may include the first support portion **141** and the second support portion **142** which extends from the first support portion **141**. The display module **150** may include the first region **151**, the second region **152**, and the third region **153** and may be disposed to surround at least a portion of the second structure **140** and configured to be movable together with the second structure. The back cover **160** may be disposed under the support member **110-3** (e.g., in the −Z-axis direction) and may form the rear surface of the electronic device **100**. Meanwhile, at least some of the components of the electronic device **100** illustrated in FIGS. **6** and **7** are identical or similar to the components of the electronic device **100** described with reference to FIGS. **1** to **5**. Therefore, repetitive descriptions will hereinafter be omitted.

(105) According to an embodiment, as described above, the electronic device **100** may be configured such that a touch input is applied to a partial region of the display module **150** visually exposed through the window region **162** in the rear direction of the electronic device **100** (e.g., the −Z-axis direction) in the first state (e.g., the state of FIG. **1** and the state of FIG. **4**A).

(106) In an embodiment, when the electronic device **100** is in the first state, a touch input may be applied to the front display region (e.g., at least a portion of the first region **151** of the display module **150**) visually exposed in the front direction of the electronic device **100** (e.g., the +Z-axis direction). Furthermore, a touch input may be applied to the rear display region (e.g., at least a portion of the second region **152** of the display module **150**) visually exposed in the rear direction of the electronic device **100** (e.g., the −Z-axis direction).

(107) In an embodiment, in the first state of the electronic device **100**, at least a portion of the second region **152** of the display module **150** may be disposed to face the window region **162** of the back cover **160**. For example, at least a portion of the second region **152** may be visually exposed on the rear surface of the electronic device **100** through the window region **162** formed to be transparent or translucent. The electronic device **100** may be configured such that a predetermined screen and/or image displayed on the second region **152** in the first state is visually exposed on the rear surface of the electronic device **100** as the window region **162** is formed in the back cover **160**. Furthermore, the electronic device **100** may be configured such that a touch input by an external input is applied to the second region **152** in the first state.

(108) According to an embodiment, a conductive layer **170** may be disposed on at least a portion of the back cover **160** to increase the sensitivity of a touch input applied to the second region **152** and improve touch recognition when the electronic device **100** is in the first state.

(109) In an embodiment, the back cover **160** may include a cover layer **161** including the window region **162** formed to be transparent or translucent and the opaque region **163**, and the conductive layer **170** disposed on at least a portion of the window region **162**.

(110) In an embodiment, the cover layer **161** may form the rear surface of the electronic device **100**. For example, a lower surface (e.g., **162***b* and **163***b*) (e.g., a surface facing the −Z-axis direction) of the cover layer **161** may be exposed outside the electronic device **100** and may form an outer surface of the electronic device **100** (e.g., the rear surface of the electronic device **100**). An upper surface (e.g., **162***a* and **163***a*) (e.g., a surface facing the +Z-axis direction) of the cover layer **161** that faces away from the lower surface (e.g., **162***b* and **163***b*) may face toward the inside of the electronic device **100**. For example, the upper surface (e.g., **162***a* and **163***a*) of the cover layer **161** may face the support member **110-3** and/or at least a portion of the display module **150**.

(111) In an embodiment, the cover layer **161** may include the window region **162** formed to be transparent or translucent and the opaque region **163** extending from the window region **162**. For

example, the cover layer **161** may be entirely formed of a transparent or translucent material, and the window region **162** and the opaque region **163** may be formed by printing a specified color or pattern on a partial region or attaching a film thereto. The cover layer **161** may be manufactured through an injection molding process. However, a manufacturing method of the cover layer **161** is not limited thereto. In another embodiment, a region in which the transparent window region **162** and the opaque region **163** are connected with each other (e.g., a boundary between the window region **162** and the opaque region **163**) may be processed in the form of a gradation, and thus the window region **162** and the opaque region **163** may be naturally connected.

(112) In an embodiment, the cover layer **161** may be formed such that the window region **162** and the opaque region **163** have the same thickness or different thicknesses. FIGS. **7**B and **7**C illustrate sectional views of the back cover **160** according to different embodiments.

(113) According to the embodiment illustrated in FIG. **7**B, the window region **162** and the opaque region **163** may have the same thickness. The upper surface **162***a* of the window region **162** and the upper surface **163***a* of the opaque region **163** may form substantially the same plane (e.g., may be coplanar with each other). The lower surface **162***b* of the window region **162** and the lower surface **163***b* of the opaque region **163** may form substantially the same plane. The conductive layer **170** may be disposed on the upper surface **162***a* of the window region **162**. The conductive layer **170** may further protrude in the +Z-axis direction by a predetermined height relative to the upper surface **163***a* of the opaque region **163**. The conductive layer **170** may formed a step with the opaque region **163** of the back cover **160**

(114) According to the embodiment illustrated in FIG. **7**C, the window region **162** and the opaque region **163** may have different thicknesses and may be formed to have a step accordingly. For example, the window region **162** may be formed to be thinner than the opaque region **163**. The upper surface **162***a* of the window region **162** and the upper surface **163***a* of the opaque region **163** may be connected with each other to form a step. The lower surface **162***b* of the window region **162** and the lower surface **163***b* of the opaque region **163** may form substantially the same plane. The conductive layer **170** may be disposed on the upper surface **162***a* of the window region **162**. The thickness of the conductive layer **170** may be substantially the same as a difference in thickness between the window region **162** and the opaque region **163**. The upper conductive surface of the conductive layer **170** may form substantially the same plane as the upper surface **163***a* of the opaque region **163**.

(115) In an embodiment, the conductive layer **170** may be disposed between the back cover **160** (e.g., the window region **162**) and the display module **150** (e.g., the second region **152**) in the first state. For example, the conductive layer **170** between the window region **162** of the back cover **160** and the second region **152** of the display module **150** may perform a function of effectively transferring, to the second region **152**, a change in permittivity caused by an external input unit (e.g., a finger or a stylus pen, or shown as a hand in FIG. **6**B) brought into contact with the window region **162**. In another embodiment, the conductive layer **170** may be formed as a partial region of the cover layer **161**. For example, the conductive layer **170** may be disposed inside the cover layer **161** without distinction of the conductive layer **170** and the cover layer **161** and thus may form one of a plurality of layers included in the cover layer **161**. In the other embodiment, the conductive layer **170** may be located inside the back cover **160** without being exposed outside the back cover **160**.

(116) In an embodiment, the conductive layer **170** may include a patterned conductive material. For example, the conductive layer **170** may include at least one electrode pattern (e.g., an electrode pattern **171**, **173**, or **175** of FIG. **10**). The electrode pattern **171**, **173**, or **175** may be formed in various forms and/or shapes to correspond to a touch pattern of the display module **150** (e.g., a pattern of driving electrodes **222** and/or a pattern of sensing electrodes **224** of FIG. **10**). The electrode pattern **171**, **173**, or **175** of the conductive layer **170** and the touch pattern of the display module **150** will be described below with reference to FIGS. **9** and **10**.

(117) In an embodiment, the conductive layer **170** may be disposed on the back cover **160**, at the window region **162**. For example, the conductive layer **170** may be disposed on the upper surface (e.g., **162***a* and **163***a*) (e.g., a surface facing the +Z-axis direction) of the cover layer **161** to face toward the inside of the electronic device **100**. The conductive layer **170** may be located to at least partially overlap the window region **162**. In another embodiment, the conductive layer **170** may be located to overlap a portion of the window region **162** and a portion of the opaque region **163**. In an embodiment, the conductive layer **170** may face at least a portion of the display module **150**. Furthermore, although not illustrated, the conductive layer **170** may face the opening region of the support member **110-3** (e.g., the opening region **111** of FIG. **3**) or at least a portion of the circuit board (e.g., the circuit board **190** of FIG. **3**) (e.g., refer to FIG. **3**). In an embodiment, the conductive layer **170** may be formed to be transparent or translucent so as not to be visually exposed through the window region **162**. For example, the conductive layer **170** may include an electrode formed of a transparent or translucent material.
(118) In the electronic device **100** according to an embodiment of the disclosure, the conductive layer **170** may be disposed between the back cover **160** and the display module **150** to improve the touch sensitivity of an external touch input applied through the back cover **160** and touch recognition when the electronic device **100** is in the first state (e.g., the state of FIG. **1**).
(119) In an embodiment, the conductive layer **170** and the second region **152** of the display module **150** may be disposed between the window region **162** of the back cover **160** and the second support portion **142** of the second structure **140**. In this case, an adhesive member B may be disposed between the second support portion **142** and the second region **152**, and an air gap (e.g., 'air gap' in FIG. **6**B) may be formed between the second region **152** and the conductive layer **170**. The air gap may be provided to secure a space in which the second structure **140** and the display module **150** are able to slide when the electronic device **100** is changed between the first state (e.g., the state of FIG. **1** and the state of FIG. **4**A) and the second state (e.g., the state of FIG. **2** and the state of FIG. **4**B). When the display module **150** moves through the air gap, a surface of the display module **150** may be prevented from being scratched or damaged by other components of the electronic device **100**, owing to the air gap.
(120) According to an embodiment, when the electronic device **100** is in the first state, the physical distance between an external input unit (e.g., a finger or a stylus pen) and the display module **150** may lower the sensitivity of a touch input. For example, the first distance d**1** between the second region **152** and the external input unit may be implemented to be slightly large due to the air gap and the back cover **160** that are disposed under the second region **152** of the display module **150** (e.g., in the −Z-axis direction), and therefore the sensitivity of the touch input by the external input unit may be lowered.
(121) In an embodiment, the conductive layer **170** disposed between the air gap and the window region **162** may be an intermediate medium capable of transferring a change in permittivity and may compensate for a decrease in the sensitivity with which the second region **152** senses an external input unit (e.g., a finger or a stylus pen).
(122) In an embodiment, to transfer a change in the permittivity of an external input unit, the conductive layer **170** may serve to increase capacitance between an electrode (e.g., a TX electrode and/or an RX electrode) included in a touch panel (e.g., the touch panel **220** of FIG. **10**) of the second region **152** and the external input unit. For example, when the external input unit is brought into contact with the back cover **160**, the amount of charge may be changed while charges are introduced into the conductive layer **170** (e.g., a patterned conductive material). At this time, the change in the amount of charge in the conductive layer **170** may be sensed by the touch panel of the display module **150** (e.g., the second region **152**). Accordingly, the first distance d**1** between the second region **152** and the external input unit may be substantially decreased to the second distance d**2** between the second region **152** and the conductive layer **170**. Thus, the sensitivity of the touch panel may be improved due to the increase in the capacitance between the touch panel and the

external input unit.
(123) The electronic device **100** according to an embodiment may increase the sensitivity of a touch sensor for each region (e.g., the first region **151**, the second region **152**, or the third region **153**) of the display module **150** depending on states of the electronic device **100**. The processor of the electronic device **100** (e.g., the processor **320** of FIG. **11**) may increase the touch sensitivity of the display module **150**, based on operational states (e.g., the first state and the second state) of the electronic device **100** and/or the direction of the electronic device **100** (e.g., when the user looks at the front surface of the electronic device **100** and when the user looks at the rear surface of the electronic device **100**).
(124) According to an embodiment, when the user looks at the rear surface of the electronic device **100** from above in the first state, the processor (e.g., the processor **320** of FIG. **11**) may recognize that the rear surface of the electronic device **100** is rotated to face a direction opposite to the direction of the earth's magnetic field, based on a rotated state of the electronic device **100** sensed through the sensor module (e.g., the sensor module **376** of FIG. **11** or a gyro sensor). In this case, the processor **320** may increase the touch sensitivity of the second region **152** visually exposed on the rear surface of the electronic device **100** through the window region **162**, thereby improving the sensitivity of a touch input through the second region **152**.
(125) According to an embodiment, the third region **153** of the display module **150** may be configured to enable a touch input. For example, in the first state, a touch input may be applied to the second region **152** through the window region **162**, and in the second state, a touch input may be applied to the third region **153** through the window region **162**. In the above embodiment, when the electronic device **100** is changed from the first state to the second state, the processor (e.g., the processor **320** of FIG. **11**) may detect the current state (e.g., the first state or the second state) of the electronic device **100** through movement of the second structure **140**, the roller members (e.g., **132** and **134**), and/or the belt member **136**, may determine a region of the display module **150** (e.g., the second region **152** or the third region **153**) located on the window region **162**, and may increase the sensitivity of a touch sensor for the corresponding region. For example, when it is detected that the electronic device **100** is currently in the first state, the processor **320** may increase the touch sensitivity of the second region **152** facing the window region **162**. Furthermore, when it is detected that the electronic device **100** is currently in the second state, the processor **320** may increase the touch sensitivity of the third region **153** facing the window region **162**.
(126) In the first state, the electronic device **100** according to an embodiment may separately display a user interface (UI) displayed on a display region (e.g., the first region **151**) visually exposed on the front surface of the electronic device **100** and a UI displayed on a display region (e.g., the second region **152**) visually exposed on the rear surface of the electronic device **100** through the window region **162**. For example, in the first state, the UI displayed on the display region (e.g., the second region **152**) exposed through the window region **162** may be implemented with a UI having a lower touch sensitivity than the UI displayed on the display region (e.g., the first region **151**) visually exposed on the front surface of the electronic device **100**. In another example, an icon (as an image) displayed on the display region visually exposed through the window region **162** may be displayed to be larger than an icon displayed through the display region visually exposed on the front surface of the electronic device **100**.
(127) FIGS. **8**A, **8**B and **8**C (otherwise referred to as FIG. **8**) illustrate the back cover of the electronic device according to an embodiment. FIGS. **8**B and **8**C illustrate a section taken along line C-C′ of FIG. **8**A.
(128) Referring to FIG. **8**, the back cover **160** may include the cover layer **161** including the window region **162** and the opaque region **163**, and the conductive layer **170** disposed on at least a portion of the window region **162**. FIG. **8** is a view illustrating an embodiment in which the conductive region **165** is added to the back cover **160** according to the embodiment of FIG. **7**. Hereinafter, repetitive descriptions will be omitted.

(129) In an embodiment, the conductive layer **170** may be configured to be electrically connected with the circuit board (e.g., the circuit board **190** of FIG. **3**).
(130) In an embodiment, the conductive region **165** electrically connected with the conductive layer **170** may be formed on the opaque region **163** of the cover layer **161** (e.g., FIG. **7**B). The conductive region **165** may be considered a part of the conductive layer **170**, such that the conductive layer **170** is extended across a boundary between the window region **162** and the opaque region **163** of the back cover **160**. The conductive region **165** may be further electrically connected to the circuit board (e.g., the circuit board **190** of FIG. **3**). As described above, the opening region (e.g., the opening region **111** of FIG. **3**) may be formed in the support member (e.g., the support member **110-3** of FIGS. **3**, **4**, and **6**), and the circuit board **190** and the back cover **160** may be disposed to face each other through the opening region **111** (e.g., refer to FIG. **3**). For example, a connecting member (e.g., a c-clip or a pogo-pin) for electrical connection may be disposed on the circuit board **190**. The connecting member may make contact with the conductive region **165**, and thus the conductive layer **170** and the circuit board **190** may be electrically connected through the conductive region **165**. In another example, a connecting member (e.g., a c-clip or a pogo-pin) for electrical connection with the circuit board **190** may be located on the conductive region **165**.
(131) In an embodiment, the conductive region **165** may be connected to a ground (GND) region of the circuit board (e.g., the circuit board **190** of FIG. **3**). In another embodiment, a specified potential may be connected to the conductive region **165** to determine the potential of the conductive layer **170**. The electronic device **100** may reduce an influence of noise and may stably operate, by electrically connecting the conductive layer **170** and the circuit board **190** through the conductive region **165**. However, the conductive region **165** does not correspond to an essential component, and according to an embodiment to which the disclosure is applied, the conductive region **165** may be omitted in a case in which the performance of the conductive layer **170** is secured even though the conductive region **165** does not exist.
(132) In an embodiment, the conductive region **165** may be integrally formed with the conductive layer **170**. For example, at least a partial region of the conductive layer **170** (e.g., the electrode pattern **173** or **175** of FIG. **9**) may extend toward the opaque region **163** to form the conductive region **165**. However, the disclosure is not limited thereto. The conductive region **165** may be formed to be a component separate from the conductive layer **170**, and a connecting member may be disposed between the conductive region **165** and the conductive layer **170** to electrically connect the conductive region **165** and the conductive layer **170**.
(133) As described above with reference to FIG. **7**, the cover layer **161** may be formed such that the window region **162** and the opaque region **163** have the same thickness or different thicknesses. FIGS. **8**B and **8**C illustrate sectional views of the back cover **160** along line C-C′ in FIG. **8**A, according to different embodiments.
(134) In an embodiment, in the case in which the window region **162** and the opaque region **163** have the same thickness, the conductive layer **170** may be disposed on the upper surface **162***a* of the window region **162**, and the conductive region **165** may be disposed on the upper surface **163***a* of the opaque region **163**. For example, the conductive layer **170** and the conductive region **165** may further protrude in the +Z-axis direction by a predetermined height beyond the upper surface **163***a* of the opaque region **163** (e.g., refer to FIG. **8**B).
(135) In an embodiment, in a case in which the window region **162** is thinner than the opaque region **163**, a step **163***c* may be formed on at least a portion of the opaque region **163** (e.g., a portion where the conductive region **165** is located). For example, the step **163***c* may be formed on a portion of the upper surface **163***a* of the opaque region **163** to correspond to the position in which the conductive region **165** is disposed.
(136) The conductive layer **170** may be disposed on the upper surface **162***a* of the window region **162**, and the conductive region **165** may be disposed on the step **163***c* of the opaque region **163**

(e.g., refer to FIG. **8**C). The conductive layer **170** and the conductive region **165** may form substantially the same plane as the upper surface **163***a* of the opaque region **163**. In the illustrated embodiment, the conductive region **165** may have substantially the same thickness as the conductive layer **170**. However, the thickness of the conductive region **165** is not limited to the illustrated embodiment, and the conductive region **165** may be formed to be thicker or thinner than the conductive layer **170**.

(137) FIGS. **9**A, **9**B, **9**C, **9**D, **9**E, **9**F and **9**G (otherwise referred to as FIG. **9**) illustrate the electrode pattern of the conductive layer of the electronic device according to an embodiment. FIGS. **9**B, **9**C, **9**D and **9**D are enlarged views of region C shown in FIG. **9**A.

(138) Referring to FIG. **9**, the conductive layer **170** of the electronic device according to an embodiment (e.g., the electronic device **100** of FIGS. **1** to **6**) may include the electrode pattern **171**, **173**, or **175**.

(139) In an embodiment, the electrode pattern **171**, **173**, or **175** of the conductive layer **170** may be formed in various forms of patterns. For example, the electrode pattern **171**, **173**, or **175** may be formed in a regular pattern, or may be formed in an irregular pattern. In FIG. **9**, the solid lines of the patterns may be solid portions of the conductive layer **170** which are spaced apart from each other along the conductive layer **170**, without being limited thereto.

(140) In an embodiment, the electrode pattern **171**, **173**, or **175** may be formed in a mesh pattern. For example, the mesh pattern may include at least one of a polygonal shape (e.g., a triangular shape like in FIG. **9**D, a rectangular shape, a square shape like in FIG. **9**B, a rhombic shape like in FIG. **9**E, a pentagonal shape, a hexagonal shape, or an octagonal shape), a circular shape like in FIG. **9**C, or a mesh shape. However, the disclosure is not limited thereto, and according to an embodiment to which the disclosure is applied, the form of the mesh pattern may be diversely modified.

(141) In an embodiment, the mesh pattern may be formed in a form in which a plurality of figures overlap or are concentric with one another (e.g., refer to FIG. **9**A). For example, the mesh pattern may include a first figure (e.g., outermost pattern) having a specified form and a plurality of figures (e.g., a plurality of inner patterns) disposed in the first figure. In this case, the plurality of figures disposed in the first figure may have a decreasing size toward the inside of the first figure and may be spaced apart from one another by a predetermined gap or different gaps.

(142) In an embodiment, a first quadrangular pattern may be disposed, and a second quadrangular pattern smaller than the first quadrangular pattern may be disposed in the first quadrangular pattern. In addition, a third quadrangular pattern smaller than the second quadrangular pattern may be disposed in the second quadrangular pattern. In another embodiment, the plurality of figures may have a pyramid shape (e.g., a triangular pyramid, a quadrangular pyramid, a pentagonal pyramid, a hexagonal pyramid, an octagonal pyramid, or a circular pyramid). The number and shape of figures disposed to overlap one another may be diversely modified. The mesh pattern having the above-described form may form predetermined points to concentrate charges transferred through the conductive layer **170** toward the inside of the figures constituting the mesh pattern, thereby further improving the sensitivity of a touch input received through the conductive layer **170** and the window region (e.g., the window region **162** of FIGS. **5** and **6**).

(143) In an embodiment, the electrode pattern **173** or **175** may include at least one of a straight line, a curved line, or a closed curve formed of a straight line or a curved line.

(144) In an embodiment, the electrode pattern **173** or **175** may be formed in a grid form in which a plurality of straight lines intersect one another at a right angle (e.g., refer to FIG. **9**F). In this case, a mesh pattern having a form in which a plurality of figures overlap one another may be disposed at points where the plurality of straight lines intersect one another. For example, the mesh patterns illustrated in FIG. **9**A may be configured to be connected with together.

(145) In an embodiment, the electrode pattern **175** may be formed in a form in which a plurality of straight lines and a plurality of closed curves intersect one another (e.g., refer to FIG. **9**G). For

example, the electrode pattern **175** may be formed in a form in which closed curves (e.g., circles) surround points where a plurality of straight lines intersect one another in a pattern having a grid form.

(146) In an embodiment, the electrode pattern **171**, **173**, or **175** may include a metal electrode pattern (e.g., a metal mesh pattern) having a mesh shape. A metal that forms the metal mesh pattern may be a metal having electrical conductivity, and the type of metal is not particularly limited. For example, a conductive metal, such as Pt, Ru, Au, Ag, Mo, Al, W, Pd, Mg, Ni, Nd, Ir, Cr, Ti, or Cu, or a metal alloy may be used as a metallic material. According to an embodiment, the electrode pattern **171**, **173**, or **175** may be implemented using silver (Ag) or copper (Cu) transparent ink.

(147) In an embodiment, the electrode pattern **171**, **173**, or **175** may include an electrode formed of a transparent or translucent material. The electrode pattern **171**, **173**, or **175** formed to be transparent or translucent may be implemented using a transparent electrode material known in the art, and the material of the electrode is not particularly limited. For example, the electrode pattern **171**, **173**, or **175** may be implemented using a transparent conductive material including indium tin oxide (ITO), indium zinc oxide (IZO), zinc oxide (ZnO), indium zinc tin oxide (IZTO), cadmium tin oxide (CTO), poly(3,4-ethylenedioxythiophene) (PEDOT), Al-doped ZnO (AZO), carbon nano tubes (CNT), graphene, and/or metal wires (e.g., silver nano wires (AGNW)). A metal used in the metal wires is not particularly limited, and these may be used alone or in combination of two or more. Alternatively, the electrode pattern **171**, **173**, or **175** may include an electrode pattern formed of ITO. In an embodiment, the electrode pattern **171**, **173**, or **175** may be implemented using various methods known in the art. The electrode pattern **171**, **173**, or **175** may be formed by various thin film deposition technologies including physical vapor deposition (PVD) or chemical vapor deposition (CVD). For example, the electrode pattern **171**, **173**, or **175** may be formed by reactive sputtering that is an example of the physical vapor deposition. Alternatively, the electrode pattern **171**, **173**, or **175** may be formed by photolithography.

(148) Furthermore, the electrode pattern **171**, **173**, or **175** may be formed through various types of printing processes. For example, various printing methods including gravure off set, reverse off set, screen printing, or gravure printing may be used. When the electrode pattern **171**, **173**, or **175** is formed by the above printing processes, the electrode pattern **171**, **173**, or **175** may be formed using a printable paste material.

(149) According to an embodiment, when the conductive layer **170** and/or the electrode pattern **171**, **173**, or **175** is formed to be thick, the permittivity may increase, and thus an effect of improving the sensitivity of a touch input may be increased.

(150) FIGS. **10**A, **10**B, **10**C, **10**D and **10**E (otherwise referred to as FIG. **10**) illustrate the display module of the electronic device according to an embodiment. FIG. **10**B illustrates a section taken along line D-D′ of FIG. **10**A.

(151) Referring to FIG. **10**, the display module **150** of the electronic device **100** according to an embodiment may include a window panel **210**, the touch panel **220**, and the display panel **230**.

(152) The embodiment of FIG. **10** is merely an example, and a configuration of the display module **150** is not necessarily limited to the illustrated embodiment. According to an embodiment to which the disclosure is applied, the display module **150** may be implemented using various types and forms of displays.

(153) In an embodiment, the window panel **210** may form an outer surface of the display module **150** and may protect the touch panel **220** or the display panel **230**. The window panel **210** may be formed of at least one of glass, poly carbonate (PC), poly methyl meth acrylate (PMMA), or polyimide (PI), or a combination of two or more thereof.

(154) In an embodiment, the touch panel **220** may be disposed facing the window panel **210**, between the window panel **210** and the display panel **230**. The touch panel **220** may be formed of a light transmitting material and may transmit an image signal displayed through the display panel **230**.

(155) In an embodiment, the touch panel **220** may sense the position, intensity, and/or duration of a touch input by using capacitance varying depending on an external input such as a user's touch input. The touch panel **220** may include a first layer **220-1**, a second layer **220-2**, and an insulating layer **220-3** located between the first layer **220-1** and the second layer **220-2**. However, the disclosure is not necessarily limited to the illustrated embodiment, and the touch panel **220** may be integrally formed without being divided into separate layers.
(156) In an embodiment, each of the first layer **220-1** and the second layer **220-2** may include a touch pattern for recognizing a touch operation of the user. For example, the touch pattern may be formed by the plurality of driving electrodes **222** and the plurality of sensing electrodes **224**.
(157) In an embodiment, the first layer **220-1** may include the pattern of the driving electrodes **222** (e.g., TX electrodes). The second layer **220-2** may include the pattern of the sensing electrodes **224** (e.g., RX electrodes). For example, the driving electrodes **222** and the sensing electrodes **224** may be disposed to cross each other in directions perpendicular to each other. The driving electrodes **222** and the sensing electrodes **224** may be disposed to cross each other in a net form, a grid form, or a matrix form. However, the form of the touch pattern (e.g., the pattern of the driving electrodes **222** and the pattern of the sensing electrodes **224**) is not necessarily limited thereto, and the touch pattern by the driving electrodes **222** and the sensing electrodes **224** may be modified in various forms.
(158) Meanwhile, without being necessarily limited to the illustrated embodiment, the touch panel **220** may be diversely modified according to an embodiment to which the disclosure is applied. For example, the first layer **220-1** may include the sensing electrodes **224**, and the second layer **220-2** may include the driving electrodes **222**.
(159) In an embodiment, the display panel **230** may be disposed between the touch panel **220** and the second structure (e.g., the second structure **140** of FIGS. **3**, **4**, and **6**). The display panel **230** may display an image in response to an electrical signal inside the electronic device **100**. The display panel **230** may be implemented with a liquid crystal display (LCD) panel or an organic light emitting diode (OLED) panel.
(160) Meanwhile, although not illustrated, an adhesive layer (not illustrated) may be disposed between the panels (e.g., the window panel **210**, the touch panel **220**, and the display panel **230**). In an embodiment, the adhesive layer may bond the panels such that the panels are not separated from each other. For example, the adhesive layer may include an optical clear adhesive (OCA) and/or a synthetic resin.
(161) Hereinafter, an interrelation of the electrode pattern (e.g., the electrode pattern **171**, **173**, or **175** of FIG. **9**) and the touch pattern (e.g., the pattern of the driving electrodes **222** and/or the pattern of the sensing electrodes **224**) will be described with reference to FIG. **9** described above.
(162) Referring to FIGS. **9** and **10**, the conductive layer **170** of the electronic device according to an embodiment (e.g., the electronic device **100** of FIGS. **1** to **6**) may include the electrode pattern **171**, **173**, or **175**. Furthermore, the display module **150** of the electronic device **100** may include the touch panel **220** in which the touch pattern (e.g., the pattern of the driving electrodes **222** and/or the pattern of the sensing electrodes **224**) is formed.
(163) According to an embodiment of the disclosure, the electronic device **100** may be configured such that the electrode pattern **171**, **173**, or **175** of the conductive layer **170** is at least partially aligned with the touch pattern (e.g., the pattern of the driving electrodes **222** and/or the pattern of the sensing electrodes **224**) of the display module **150** when the electronic device **100** is in the first state (e.g., the state of FIG. **1** and the state of FIG. **4**A).
(164) For example, the electrode pattern **171**, **173**, or **175** may be formed in a form at least partially corresponding to the touch pattern of the touch panel **220**. In this case, in the first state, the touch panel **220** of the second region **152** may be disposed to face the conductive layer **170**, and thus the touch pattern and the electrode pattern **171**, **173**, or **175** may be configured to at least partially correspond to each other. For example, points (e.g., touch input recognition points or touch nodes)

at which a plurality of touch electrodes (e.g., the driving electrodes **222** and the sensing electrodes **224**) intersect in the touch panel **220** may be disposed to at least partially coincide with predetermined points or intersecting points concentrated in the direction toward the inside of the electrode pattern **171**, **173**, or **175** of the conductive layer **170**. According to an embodiment, the density of the electrode pattern **171**, **173**, or **175** may be lower than the density of the touch pattern (e.g., the pattern of the driving electrodes **222** and/or the pattern of the sensing electrodes **224**).
(165) By partially corresponding the electrode pattern **171**, **173**, or **175** of the conductive layer **170** and the touch input recognition points (e.g., touch nodes) at which the plurality of touch electrodes (e.g., the driving electrodes **222** and the sensing electrodes **224**) intersect in the touch panel **220**, a change in the amount of charge in the electrode pattern **171**, **173**, or **175** may be more effectively transferred to the touch input recognition points, and thus an effect of improving the sensitivity and accuracy of a touch input may be further increased.
(166) FIG. **11** is a block diagram illustrating an electronic device in a network environment according to an embodiment.
(167) Referring to FIG. **11**, the electronic device **301** in the network environment **300** may communicate with an electronic device **302** via a first network **398** (e.g., a short-range wireless communication network), or an electronic device **304** or a server **308** via a second network **399** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **301** may communicate with the electronic device **304** via the server **308**. According to an embodiment, the electronic device **301** may include a processor **320**, memory **330**, an input device **350**, a sound output device **355**, a display device **360**, an audio module **370**, a sensor module **376**, an interface **377**, a haptic module **379**, a camera module **380**, a power management module **388**, a battery **389**, a communication module **390**, a subscriber identification module (SIM) **396**, or an antenna module **397**. In some embodiments, at least one (e.g., the display device **360** or the camera module **380**) of the components may be omitted from the electronic device **301**, or one or more other components may be added in the electronic device **301**. In some embodiments, some of the components may be implemented as single integrated circuitry. For example, the sensor module **376** (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be implemented as embedded in the display device **360** (e.g., a display).
(168) The processor **320** may execute, for example, software (e.g., a program **340**) to control at least one other component (e.g., a hardware or software component) of the electronic device **301** coupled with the processor **320**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **320** may load a command or data received from another component (e.g., the sensor module **376** or the communication module **390**) in volatile memory **332**, process the command or the data stored in the volatile memory **332**, and store resulting data in non-volatile memory **334**. According to an embodiment, the processor **320** may include a main processor **321** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **323** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **321**. Additionally or alternatively, the auxiliary processor **323** may be adapted to consume less power than the main processor **321**, or to be specific to a specified function. The auxiliary processor **323** may be implemented as separate from, or as part of the main processor **321**.
(169) The auxiliary processor **323** may control at least some of functions or states related to at least one component (e.g., the display device **360**, the sensor module **376**, or the communication module **390**) among the components of the electronic device **301**, instead of the main processor **321** while the main processor **321** is in an inactive (e.g., sleep) state, or together with the main processor **321** while the main processor **321** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **323** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **380** or the

communication module **390**) functionally related to the auxiliary processor **323**.
(170) The memory **330** may store various data used by at least one component (e.g., the processor **320** or the sensor module **376**) of the electronic device **301**. The various data may include, for example, software (e.g., the program **340**) and input data or output data for a command related thereto. The memory **330** may include the volatile memory **332** or the non-volatile memory **334**.
(171) The program **340** may be stored in the memory **330** as software, and may include, for example, an operating system (OS) **342**, middleware **344**, or an application **346**.
(172) The input device **350** may receive a command or data to be used by other component (e.g., the processor **320**) of the electronic device **301**, from the outside (e.g., a user) of the electronic device **301**. The input device **350** may include, for example, a microphone, a mouse, a keyboard, or a digital pen (e.g., a stylus pen).
(173) The sound output device **355** may output sound signals to the outside of the electronic device **301**. The sound output device **355** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for an incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.
(174) The display device **360** may visually provide information to the outside (e.g., a user) of the electronic device **301**. The display device **360** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display device **360** may include touch circuitry adapted to detect a touch, or sensor circuitry (e.g., a pressure sensor) adapted to measure the intensity of force incurred by the touch.
(175) The audio module **370** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **370** may obtain the sound via the input device **350**, or output the sound via the sound output device **355** or a headphone of an external electronic device (e.g., an electronic device **302**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **301**.
(176) The sensor module **376** may detect an operational state (e.g., power or temperature) of the electronic device **301** or an environmental state (e.g., a state of a user) external to the electronic device **301**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **376** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
(177) The interface **377** may support one or more specified protocols to be used for the electronic device **301** to be coupled with the external electronic device (e.g., the electronic device **302**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **377** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(178) A connecting terminal **378** may include a connector via which the electronic device **301** may be physically connected with the external electronic device (e.g., the electronic device **302**). According to an embodiment, the connecting terminal **378** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).
(179) The haptic module **379** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **379** may include, for example, a motor, a piezoelectric element, or an electric stimulator.
(180) The camera module **380** may capture a still image or moving images. According to an embodiment, the camera module **380** may include one or more lenses, image sensors, image signal

processors, or flashes.
(181) The power management module **388** may manage power supplied to the electronic device **301**. According to one embodiment, the power management module **388** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).
(182) The battery **389** may supply power to at least one component of the electronic device **301**. According to an embodiment, the battery **389** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.
(183) The communication module **390** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **301** and the external electronic device (e.g., the electronic device **302**, the electronic device **304**, or the server **308**) and performing communication via the established communication channel. The communication module **390** may include one or more communication processors that are operable independently from the processor **320** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **390** may include a wireless communication module **392** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **394** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **398** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **399** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **392** may identify and authenticate the electronic device **301** in a communication network, such as the first network **398** or the second network **399**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **396**.
(184) The antenna module **397** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **301**. According to an embodiment, the antenna module **397** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., PCB). According to an embodiment, the antenna module **397** may include a plurality of antennas. In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **398** or the second network **399**, may be selected, for example, by the communication module **390** (e.g., the wireless communication module **392**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **390** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **397**.
(185) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).
(186) According to an embodiment, commands or data may be transmitted or received between the electronic device **301** and the external electronic device **304** via the server **308** coupled with the second network **399**. Each of the electronic devices **302** and **304** may be a device of a same type as, or a different type, from the electronic device **301**. According to an embodiment, all or some of operations to be executed at the electronic device **301** may be executed at one or more of the

external electronic devices **302**, **304**, or **308**. For example, if the electronic device **301** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **301**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **301**. The electronic device **301** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

(187) FIG. **12** is a block diagram illustrating the display device according to an embodiment.

(188) Referring to FIG. **12**, the display device **360** may include a display **410** and a display driver integrated circuit (DDI) **430** to control the display **410**. The DDI **430** may include an interface module **431**, memory **433** (e.g., buffer memory), an image processing module **435**, or a mapping module **437**. The DDI **430** may receive image information that contains image data or an image control signal corresponding to a command to control the image data from another component of the electronic device **301** via the interface module **431**. For example, according to an embodiment, the image information may be received from the processor **320** (e.g., the main processor **321** (e.g., an application processor)) or the auxiliary processor **323** (e.g., a graphics processing unit) operated independently from the function of the main processor **321**. The DDI **430** may communicate, for example, with touch circuitry **450** or the sensor module **376** via the interface module **431**. The DDI **430** may also store at least part of the received image information in the memory **433**, for example, on a frame by frame basis. The image processing module **435** may perform pre-processing or post-processing (e.g., adjustment of resolution, brightness, or size) with respect to at least part of the image data. According to an embodiment, the pre-processing or post-processing may be performed, for example, based at least in part on one or more characteristics of the image data or one or more characteristics of the display **410**. The mapping module **437** may generate a voltage value or a current value corresponding to the image data pre-processed or post-processed by the image processing module **435**. According to an embodiment, the generating of the voltage value or current value may be performed, for example, based at least in part on one or more attributes of the pixels (e.g., an array, such as an RGB stripe or a pentile structure, of the pixels, or the size of each subpixel). At least some pixels of the display **410** may be driven, for example, based at least in part on the voltage value or the current value such that visual information (e.g., a text, an image, or an icon) corresponding to the image data may be displayed via the display **410**.

(189) According to an embodiment, the display device **360** may further include the touch circuitry **450**. The touch circuitry **450** may include a touch sensor **451** and a touch sensor IC **453** to control the touch sensor **451**. The touch sensor IC **453** may control the touch sensor **451** to sense a touch input or a hovering input with respect to a certain position on the display **410**. To achieve this, for example, the touch sensor **451** may detect (e.g., measure) a change in a signal (e.g., a voltage, a quantity of light, a resistance, or a quantity of one or more electric charges) corresponding to the certain position on the display **410**. The touch circuitry **450** may provide input information (e.g., a position, an area, a pressure, or a time) indicative of the touch input or the hovering input detected via the touch sensor **451** to the processor **320**. According to an embodiment, at least part (e.g., the touch sensor IC **453**) of the touch circuitry **450** may be formed as part of the display **410** or the DDI **430**, or as part of another component (e.g., the auxiliary processor **323**) disposed outside the display device **360**.

(190) According to an embodiment, the display device **360** may further include at least one sensor (e.g., a fingerprint sensor, an iris sensor, a pressure sensor, or an illuminance sensor) of the sensor module **376** or a control circuit for the at least one sensor. In such a case, the at least one sensor or the control circuit for the at least one sensor may be embedded in one portion of a component (e.g.,

the display **410**, the DDI **430**, or the touch circuitry **450**)) of the display device **360**. For example, when the sensor module **376** embedded in the display device **360** includes a biometric sensor (e.g., a fingerprint sensor), the biometric sensor may obtain biometric information (e.g., a fingerprint image) corresponding to a touch input received via a portion of the display **410**. As another example, when the sensor module **376** embedded in the display device **360** includes a pressure sensor, the pressure sensor may obtain pressure information corresponding to a touch input received via a partial or whole area of the display **410**. According to an embodiment, the touch sensor **451** or the sensor module **376** may be disposed between pixels in a pixel layer of the display **410**, or over or under the pixel layer.

(191) An electronic device **100** according to an embodiment of the disclosure may include a first structure **120**, a second structure **140** that is connected to the first structure **120** so as to slide relative to the first structure **120** and that surrounds at least a portion of the first structure **120**, a display module **150** that is disposed on the second structure **140** and that is movable together with the second structure **140** relative to the first structure **120**, and a back cover **160** that forms at least a portion of an outer surface of the electronic device **100** and that is disposed to face the first structure **120**. The display module **150** may include a first region **151** and a second region **152** that extends from the first region **151**, and the first region **151** or the second region **152** may be formed to be at least partially flexible. The back cover **160** may include a window region **162** having at least a partial region formed of a transparent or translucent material and a conductive layer **170** disposed on at least a portion of the window region **162** to face toward the first structure **120**. The electronic device **100** may include a first state in which the first region **151** forms a front surface of the electronic device **100** and at least a portion of the second region **152** is disposed between the first structure **120** and the back cover **160** and visually exposed on a rear surface of the electronic device **100** through the window region **162** and a second state in which at least a portion of the second region **152** forms the front surface of the electronic device **100** together with the first region **151**. In the first state, at least a portion of the second region **152** may be configured to receive a touch input from the outside through the window region **162** and the conductive layer **170**.

(192) In an embodiment, an electronic device includes a display module which is flexible and displays an image, the display module including a first display region **151** which forms a front surface of the electronic device, and a second display region **152** which extends from the first display region and tactically senses an external input (e.g., a touch input) via a conductive layer, a first structure facing the display module and along which the display module is slidable, a second structure slidably connected to the first structure, the second structure being further connected to the display module and slidable together therewith along the first structure, and a back cover which faces the first structure and forms a rear surface of the electronic device which is opposite to the front surface, the back cover including a window region through which the image is viewable and the external input is sensable, and the conductive layer on the window region and facing the first structure. The second structure which slides towards the first structure closes the electronic device (e.g., a first state), the second structure which slides away from the first structure opens the electronic device (e.g., a second state), the electronic device which is closed defines the front surface including the first display region, together with disposing the second display region between the first structure and back cover and tactically exposed at the window region of the back cover, and the electronic device which is open defines the front surface including the first display region, together with a portion of the second display region.

(193) In various embodiments, the conductive layer **170** may include at least one electrode pattern **171**, **173**, or **175**, and the electrode pattern **171**, **173**, or **175** may include at least one of a straight line, a curved line, or a closed curve formed of a straight line or a curved line. That is, the electrode pattern includes at least one of a straight line, a curved line, or a closed shape (e.g., an enclosed shape like shown in FIGS. **9**B, **9**C, **9**D and **9**E).

(194) In various embodiments, the conductive layer **170** may include an electrode formed of a

transparent or translucent material.
(195) In various embodiments, the electrode pattern **171**, **173**, or **175** may include a metal electrode pattern or indium tin oxide (ITO) electrode pattern **171**, **173**, or **175** that has a mesh shape.
(196) In various embodiments, the electrode pattern **171**, **173**, or **175** may be formed in a form in which a plurality of circular or polygonal figures overlap one another. That is, the electrode pattern includes a plurality of circular shapes or a plurality of polygonal shapes which overlap one another.
(197) In various embodiments, the display module **150** may include a display panel **230** and a touch panel **220** disposed on at least one surface of the display panel **230**. The touch panel **220** may include a touch pattern formed by a plurality of driving electrodes **222** and a plurality of sensing electrodes **224**. In the first state, the display panel **230** and at least a portion of the touch panel **220** may face the conductive layer **170**, and the electrode pattern may be configured to be at least partially aligned with the touch pattern.
(198) In an embodiment, the display module further includes a display panel and a touch panel which faces the display panel, the touch panel includes a touch pattern having a plurality of driving electrodes and a plurality of sensing electrodes, and the electronic device which is closed disposes the display panel and the touch panel facing the conductive layer on the window region, together with the electrode pattern of the conductive layer aligned with the touch pattern of the touch panel.
(199) In various embodiments, the back cover **160** may include an opaque region **163** that extends from the window region **162**, and a conductive region **165** electrically connected with the conductive layer **170** may be formed on at least a portion of the opaque region **163**. That is, the back cover further includes an opaque region extended from the window region, and a conductive region which is on the opaque region and electrically connected with the conductive layer on the window region.
(200) In various embodiments, the electronic device **100** may further include a printed circuit board (PCB) **190** disposed between the first structure **120** and the back cover **160**, and the conductive region **165** may be electrically connected with the printed circuit board **190**.
(201) In various embodiments, at least a portion of the conductive layer **170** may extend toward the opaque region **163** such that the conductive region **165** is integrally formed with the conductive layer **170**. That is, the conductive layer on the window region extends toward the opaque region to define the conductive region on the opaque region.
(202) In various embodiments, the second structure **140** may include a first support portion **141** on which one portion of the first region **151** is disposed and a second support portion **142** that extends from the first support portion **141** and on which another portion of the first region **151** and the second region **152** are disposed.
(203) In an embodiment, a first portion of the first display region, a second portion of the first display region and the second display region are arranged in a first direction (e.g., along the X axis direction). The second structure includes a first support portion corresponding to the first portion of the first display region, and a second support portion which is extended from the first support portion in the first direction and corresponds to the second portion of the first display region and to the second display region.
(204) In various embodiments, at least part of the second support portion **142** may be inserted into or withdrawn from a space between the first structure **120** and the back cover **160** as the second structure **140** slides relative to the first structure **120**. That is, the first structure and the back cover provide a space therebetween. Sliding of the second structure, together with the display module, along the first structure, provides insertion of the second support portion into or withdrawal of the second support portion from the space between the first structure and the back cover.
(205) In various embodiments, the first structure **120** may include a first surface **121** and a second surface **122** that faces away from the first surface **121**, and when the electronic device **100** is changed from the first state to the second state, at least part of the second support portion **142** may be separated from between the second surface **122** of the first structure **120** and the back cover **160**

and may be disposed to face the first surface **121** of the first structure **120**. That is, the first structure includes a first surface, and a second surface which is opposite to the first surface and closer to the rear surface of the electronic device than the first surface. The electronic device which is closed disposes a portion of the second support portion between the second surface of the first structure and the back cover, and the electronic device which is open disposes the portion of the second support portion region facing the first surface of the first structure.

(206) In various embodiments, the second support portion **142** may be formed of a bendable material so as to at least partially form a curved surface in the first state or the second state. That is, the first structure includes an end along which the display module is slidable together with the second structure. The electronic device which is closed or the electronic device which is open disposes the second support portion of the second structure having a curved surface along the end of the first structure.

(207) In various embodiments, the second support portion **142** may include a first edge **142***a* connected with the first support portion **141** and a second edge **142***b* that forms an end portion of the second support portion, and the first edge **142***a* and the second edge **142***b* may be configured to move in opposite directions, respectively, when the second structure **140** slides relative to the first structure **120**.

(208) In an embodiment, the second support portion includes a first edge which is closest to the first support portion, and at which the first support portion is connected to the second support portion, and a second edge which is furthest from the first support portion and forms an end portion of the second structure. Sliding of the second structure, together with the display module, along the first structure, moves the first edge and the second edge in opposite directions along the first structure.

(209) In various embodiments, the first structure **120** may include a first roller member **132** disposed on one side of the first structure **120** so as to be rotatable, a second roller member **134** disposed on an opposite side of the first structure **120** so as to be rotatable, and a belt member **136** having at least a portion disposed to surround the second roller member **134** and opposite end portions connected to the second structure **130**. Each of the first roller member **132** and the second roller member **134** may be configured to rotate in a direction the same as a sliding direction of the second structure **140** when the second structure **140** slides.

(210) In an embodiment, the second structure is slidable together with the display module along the first structure, along a sliding direction. The first structure includes a first roller which is disposed at a first end the first structure and rotatable in the sliding direction, a second roller which is disposed at a second end opposite to the first end of the first structure and rotatable in the sliding direction, and a belt which extends along the second roller, the belt including opposite ends each connected to the second structure.

(211) In various embodiments, the second structure **140** may include a first support portion **141** and a second support portion **142** that extends from the first support portion **141**. The belt member **136** may be disposed such that one end portion **136***a* is connected to the first support portion and an opposite end portion **136***b* is connected to the second support portion **142** in a state in which the belt member **136** surrounds the second roller member **134**, and the belt member **136** may provide tension to maintain the second support portion **142** in a tight state.

(212) In an embodiment, a first portion of the first display region, a second portion of the first display region and the second display region are arranged in a first direction. The second structure includes a first support portion corresponding to the first portion of the first display region, and a second support portion which is extended from the first support portion in the first direction and corresponds to the second portion of the first display region and to the second display region. The belt provides tension to the second support portion, the belt including a first end connected to the first support portion, and a second end connected to the second support portion.

(213) In various embodiments, the second support portion **142** of the second structure **140** may be

disposed to surround at least a portion of the first roller member **132**, and the one end portion **136***a* and the opposite end portion **136***b* of the belt member **136** may be moved in opposite directions by the first roller member **132** and the second roller member **134** when the second structure **140** slides. That is, the second support portion of the second structure extends along the first roller, and sliding of the second structure, together with the display module, along the first structure, rotates the first roller and the second roller, together with moving the first end and the second end of the belt in opposite directions along the first structure.

(214) In various embodiments, when the second structure **140** slides, the first roller member **132** may rotate while making contact with the second support portion **142**, and the second roller member **134** may rotate while making contact with the belt member **136**. That is,

(215) Sliding of the second structure, together with the display module, along the first structure, disposes the second support portion in contact with the first roller to rotate the first roller, together with disposing the belt in contact with the second roller to rotate the second roller.

(216) An electronic device **100** according to an embodiment of the disclosure may include a case **110** including a support member **110-3**, a first side member **110-1**, and a second side member **110-2**, the first side member **110-1** and the second side member **110-2** being disposed on opposite end portions of the support member **110-3**, a first structure **120** having at least a portion surrounded by the case **110**, a second structure **140** connected to the first structure **120** so as to slide relative to the first structure **120**, a display module **150** disposed on the second structure **140** so as to be movable together with the second structure **140** relative to the first structure **120** and formed to be at least partially flexible, the display module **150** including a first region **151** and a second region **152** that extends from the first region **151**, and a back cover **160** that forms at least a portion of a rear surface of the electronic device **100** and that is disposed to surround the support member **110-3**, the back cover **160** including a window region **162** formed to be transparent and a conductive layer **170** disposed on at least a portion of the window region **162**. The electronic device **100** may be changed between a first state and a second state as the second structure **140** slides. In the first state, the first region **151** may form a front surface of the electronic device, **100** and at least a portion of the second region **152** may be visually exposed on the rear surface of the electronic device **100** through the window region **162**. In the second state, at least a portion of the second region **152** may form the front surface of the electronic device **100** together with the first region **151**. In the first state, at least a portion of the second region **152** may be located to face the window region **162** and the conductive layer **170** and may be configured to receive a touch input from the outside through the window region **162** and the conductive layer **170**.

(217) In an embodiment, an electronic device includes a case including a support member including a first end and a second end opposite to each other in a first direction, a first side member which is connected to the support member at the first end, and a second side member which is connected to the support member at the second end, a display module which is flexible and displays an image, the display module including a first display region which forms a front surface of the electronic device, and a second display region which extends from the first display region and tactically senses an external input via a conductive layer, a first structure facing the display module and along which the display module is slidable between the first side member and the second side member of the case and in a sliding direction which crosses the first direction, a second structure slidably connected to the first structure, the second structure being further connected to the display module and slidable together therewith along the first structure, and a back cover which faces the first structure with the support member of the case therebetween, and forms a rear surface of the electronic device which is opposite to the front surface, the back cover including a window region through which the image is viewable and the external input is sensable, and the conductive layer on the window region and facing the first structure. The second structure which slides towards the first structure closes the electronic device, the second structure which slides away from the first structure opens the electronic device, the electronic device which is closed defines the front

surface including the first display region, together with disposing the second display region between the first structure and back cover and tactically exposed at the window region of the back cover, and the electronic device which is open defines the front surface including the first display region, together with a portion of the second display region.
(218) In various embodiments, the conductive layer **170** may include at least one electrode pattern **171**, **173**, or **175**, and the electrode pattern **171**, **173**, or **175** may include a metal electrode pattern or indium tin oxide (ITO) electrode pattern having a mesh shape.
(219) The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.
(220) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. For example, a reference number labeling a singular form of an element within the drawing figures may be used to reference a plurality of the singular element within the text of specification.
(221) It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, “a”, “an,” “the,” and “at least one” do not denote a limitation of quantity, and are intended to include both the singular and plural, unless the context clearly indicates otherwise. For example, “an element” has the same meaning as “at least one element,” unless the context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. “Or” means “and/or.” As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” or “includes” and/or “including” when used in this specification, specify the presence of stated features, regions, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, regions, integers, steps, operations, elements, components, and/or groups thereof.
(222) As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order).
(223) It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly, without an intervening element, without a third element, etc.), wirelessly, or via a third element. In contrast, when an element is referred to as being “directly” related to another element, there is no intervening element or third element therebetween. As being in contact, elements may form an interface therebetween.
(224) Furthermore, relative terms, such as “lower” or “bottom” and “upper” or “top,” may be used herein to describe one element's relationship to another element as illustrated in the Figures. It will be understood that relative terms are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures. For example, if the device in one of the figures is turned over, elements described as being on the “lower” side of other elements would then be oriented on “upper” sides of the other elements. The term “lower,” can therefore, encompasses both an orientation of “lower” and “upper,” depending on the particular orientation of the figure.

Similarly, if the device in one of the figures is turned over, elements described as “below” or “beneath” other elements would then be oriented “above” the other elements. The terms “below” or “beneath” can, therefore, encompass both an orientation of above and below.
(225) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the present disclosure, and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.
(226) Embodiments are described herein with reference to cross section illustrations that are schematic illustrations of idealized embodiments. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, embodiments described herein should not be construed as limited to the particular shapes of regions as illustrated herein but are to include deviations in shapes that result, for example, from manufacturing. For example, a region illustrated or described as flat may, typically, have rough and/or nonlinear features. Moreover, sharp angles that are illustrated may be rounded. Thus, the regions illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the precise shape of a region and are not intended to limit the scope of the present claims.
(227) As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic”, “logic block”, “part”, or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).
(228) Various embodiments as set forth herein may be implemented as software (e.g., the program **340**) including one or more instructions that are stored in a storage medium (e.g., internal memory **336** or external memory **338**) that is readable by a machine (e.g., the electronic device **301**). For example, a processor (e.g., the processor **320**) of the machine (e.g., the electronic device **301**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Where, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.
(229) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
(230) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g.,

modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: a display module which is flexible and displays an image, the display module including: a first display region which forms a front surface of the electronic device, a second display region which extends from the first display region, and a touch panel which is in the first display region and the second display region and senses an external touch input; a first structure which faces the display module and along which the display module is slidable; a second structure slidably connected to the first structure, the second structure being further connected to the display module; a back cover which forms a rear surface of the electronic device, the back cover including a window region through which the second display region is visually exposed; and a conductive layer which is on the window region of the back cover, faces the second display region of the display module, into which an electrical charge is introduced by the external touch input via the window region, and within which electrical charge is changed by the electrical charge of the external touch input, wherein in response to the external touch input on the window region, the touch panel in the second display region senses the changed electrical charge through the conductive layer.
2. The electronic device of claim 1, wherein the conductive layer which is on the back cover, includes an electrode pattern, and the electrode pattern includes a straight line, a curved line, or a closed shape having a straight line or a curved line.
3. The electronic device of claim 2, wherein the electrode pattern has a mesh shape, and the electrode pattern which has the mesh shape includes a metal electrode pattern or indium tin oxide electrode pattern.
4. The electronic device of claim 2, wherein the electrode pattern includes a plurality of circular shapes or a plurality of polygonal shapes which overlap one another.
5. The electronic device of claim 2, wherein the display module further includes a display panel which faces the touch panel, the touch panel includes a touch pattern having a plurality of driving electrodes and a plurality of sensing electrodes, and the electronic device which is closed disposes the display panel and the touch panel facing the conductive layer on the window region, together with the electrode pattern of the conductive layer aligned with the touch pattern of the touch panel.
6. The electronic device of claim 1, wherein the conductive layer which is on the back cover, includes an electrode including a transparent material or translucent material.
7. The electronic device of claim 1, wherein the back cover further includes: an opaque region extended from the window region, and a conductive region which is on the opaque region and electrically connected with the conductive layer on the window region.
8. The electronic device of claim 7, wherein the conductive layer on the window region extends toward the opaque region to define the conductive region on the opaque region.
9. The electronic device of claim 1, wherein a first portion of the first display region, a second portion of the first display region and the second display region are arranged in a first direction, and the second structure includes: a first support portion corresponding to the first portion of the first display region; and a second support portion which is extended from the first support portion in the first direction and corresponds to the second portion of the first display region and to the second display region.

10. The electronic device of claim 9, wherein the first structure and the back cover provide a space therebetween, and sliding of the second structure, together with the display module, along the first structure, provides insertion of the second support portion into or withdrawal of the second support portion from the space between the first structure and the back cover.
11. The electronic device of claim 9, wherein the first structure includes a first surface, and a second surface which is opposite to the first surface and closer to the rear surface of the electronic device than the first surface, the electronic device which is closed disposes a portion of the second support portion between the second surface of the first structure and the back cover, and the electronic device which is open disposes the portion of the second support portion region facing the first surface of the first structure.
12. The electronic device of claim 9, wherein the first structure includes an end along which the display module is slidable together with the second structure, and the electronic device which is closed or the electronic device which is open disposes the second support portion of the second structure having a curved surface along the end of the first structure.
13. The electronic device of claim 9, wherein the second support portion includes: a first edge which is closest to the first support portion, and at which the first support portion is connected to the second support portion, and a second edge which is furthest from the first support portion and forms an end portion of the second structure, and sliding of the second structure, together with the display module, along the first structure, moves the first edge and the second edge in opposite directions along the first structure.
14. The electronic device of claim 1, wherein the second structure is slidable together with the display module along the first structure, along a sliding direction, and the first structure includes: a first roller which is disposed at a first end the first structure and rotatable in the sliding direction, a second roller which is disposed at a second end opposite to the first end of the first structure and rotatable in the sliding direction, and a belt which extends along the second roller, the belt including opposite ends each connected to the second structure.
15. The electronic device of claim 14, wherein a first portion of the first display region, a second portion of the first display region and the second display region are arranged in a first direction, the second structure includes: a first support portion corresponding to the first portion of the first display region; and a second support portion which is extended from the first support portion in the first direction and corresponds to the second portion of the first display region and to the second display region, and the belt provides tension to the second support portion, the belt including: a first end connected to the first support portion, and a second end connected to the second support portion.
16. The electronic device of claim 15, wherein the second support portion of the second structure extends along the first roller, and sliding of the second structure, together with the display module, along the first structure, rotates the first roller and the second roller, together with moving the first end and the second end of the belt in opposite directions along the first structure.
17. The electronic device of claim 16, wherein sliding of the second structure, together with the display module, along the first structure, disposes the second support portion in contact with the first roller to rotate the first roller, together with disposing the belt in contact with the second roller to rotate the second roller.
18. An electronic device comprising: a display module which is flexible and displays an image, the display module including: a first display region which forms a front surface of the electronic device, and a second display region which extends from the first display region and senses an external touch input via a conductive layer: a first structure which faces the display module and along which the display module is slidable; a second structure slidably connected to the first structure, the second structure being further connected to the display module; a printed circuit board disposed between the first structure and the back cover; a back cover which forms a rear surface of the electronic device, the back cover including: a window region through which the

second display region is visually exposed, and on which the conductive layer is disposed, an opaque region extending from the window region, and a conductive region which is on the opaque region and electrically connected with the conductive layer on the window region and the printed circuit board, wherein the conductive region of the back cover is electrically connected with the printed circuit board, and in response to the external touch input occurring on the window region, a change of charge of the conductive layer on the window region is transferred to at least a portion of the second display region and the at least a portion of the second display region is configured to detect the external touch input through the window region.

19. An electronic device comprising: a case including a support member including a first end and a second end opposite to each other in a first direction, a first side member which is connected to the support member at the first end, and a second side member which is connected to the support member at the second end; a display module which is flexible and displays an image, the display module including: a first display region which forms a front surface of the electronic device, a second display region which extends from the first display region, and a touch panel which is in the first display region and the second display region and senses an external touch input; a first structure which faces the display module and along which the display module is slidable between the first side member and the second side member of the case and in a sliding direction which crosses the first direction; a second structure slidably connected to the first structure, the second structure being further connected to the display module; a back cover which forms a rear surface of the electronic device, the back cover including a window region through which the second display region is visually exposed; and a conductive layer which is on the window region of the back cover, faces the second display region of the display module, into which an electrical charge is introduced by the external touch input via the window region, and within which electrical charge is changed by the electrical charge of the external touch input, wherein in response to the external touch input on the window region, the touch panel in the second display region senses the changed electrical charge through the conductive layer.

20. The electronic device of claim 19, wherein the conductive layer which is on the back cover includes an electrode pattern, the electrode pattern has a mesh shape, and the electrode pattern which has the mesh shape includes a metal electrode pattern or indium tin oxide electrode pattern.